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SPD801: Master Thesis in Computer Science

Formalising πLL in Lean

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Abstract

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1 Introduction

The Curry-Howard correspondence establishes a deep connection between logic and computation, where proofs correspond to programs and logical propositions correspond to types. This perspective has been particularly influential in the study of concurrent systems, where linear logic and session types provide a logical foundation for structured communication.

In this setting, linear logic captures resource-sensitive reasoning, while session types describe communication protocols between concurrent processes. The resulting correspondence, often referred to as propositions-as-sessions, provides a principled interpretation of interaction as logical proof transformation, and has been further related to process calculi such as the π-calculus.

This line of work has been further developed by [Montesi and Peressotti 2021], who provide a unified account reconciling linear logic, the π-calculus, and their metatheory. Their framework serves as the formal basis for the system πLL considered in this thesis.

Motivation and correctness concerns. Despite the elegance of these theoretical correspondences, their informal and paper-based presentations are prone to subtle errors. In particular, the application of inference rules in linear logic and session-typed systems often relies on implicit ordering conventions and intuitive interpretations of syntax. This can lead to mistakes that are not immediately apparent in written derivations.

As an illustrative example, small inconsistencies can arise in the presentation of typing rules within such systems, where the intuitive alignment between variable ordering and type assignment does not precisely match the formal rule structure. These discrepancies are not flaws of the underlying theory, but rather reflect the difficulty of maintaining precision in informal presentations, including in the framework of Montesi and Peressotti.

As observed in discussions with the authors of the framework, such issues are easy to overlook because rule application is often guided by intuition rather than strict syntactic structure. This highlights that even formally correct rules may be unintuitive to apply due to their representation.

These observations motivate a more disciplined approach to formal reasoning, where correctness is ensured by construction rather than by inspection.

Why mechanization. Mechanization provides a means of bridging this gap. Rather than extending the underlying theory, the goal of mechanization in this work is to validate and internalize existing results within a proof assistant. This allows us to eliminate ambiguity arising from informal presentation and to ensure that all derivations are checked with respect to a precise formal semantics.

Moreover, mechanized developments have repeatedly demonstrated their ability to uncover small but meaningful inconsistencies in paper proofs, supporting the view that proof assistants serve as a valuable tool for verifying, rather than merely implementing, theoretical frameworks.

Contributions. In this thesis, we present a mechanized formalization of the π-calculus-based linear logic system πLL as developed by Montesi and Peressotti, with particular emphasis on its multiplicative fragment. The contributions are as follows:

- A formalization of the syntax and typing system of πLL in Lean, including processes, session types, and substitution operations.
- A formalization of the operational semantics for the multiplicative fragment of πLL.
- Formal proofs of session fidelity for the multiplicative fragment.
- A locally nameless representation of binding, eliminating the need for reasoning up to α-equivalence and simplifying substitution reasoning.

Scope of the formalization. This thesis focuses on a mechanized reconstruction of the πLL system of [Montesi and Peressotti 2021], with particular emphasis on its multiplicative fragment. The

syntactic and typing components of πLL have been fully mechanized, including processes, session types, typing judgments, and substitution operations. The operational semantics are defined for the multiplicative fragment, with the exception of the res-rule.

Consequently, metatheoretic results concerning typing apply to the full system, while results involving operational semantics are restricted to the multiplicative fragment. The aim of this work is not to extend the theory, but to provide a mechanically verified account of an existing theoretical framework.

2 Background

This section begins by introducing the theoretical foundations underlying the formalisation following the presentation of [Montesi and Peressotti 2021], We cover the full πLL calculus, including processes, types, typing, environments / hyperenvironments, and judgements, then restrict attention to the multiplicative fragment πMLL when discussing operational semantics.

Following [Montesi and Peressotti 2021], we use *blue* to highlight types and *red* for process terms throughout this thesis.

2.1 Classical Linear Logic, CLL

Linear logic [Girard 1987] is a refinement of classical and intuitionistic logic. In contrast to these systems, where formulas can be freely duplicated or discarded, linear logic restricts structural rules such as weakening and contraction, and instead treats formulas as consumable resources. This yields a resource-sensitive interpretation of logic, where each assumption must be used in a controlled manner [Di Cosmo and Miller 2023]. In this thesis, we work in the classical presentation of linear logic, where sequents are symmetric and duality replaces the distinction between assumptions and conclusions.

In particular, conjunction and disjunction split into multiplicative and additive variants. The multiplicative fragment forms the core of the system and includes the tensor connective $A \otimes B$ and its dual $A \parr B$, together with the corresponding units 1 and $\bot$. These connectives capture composition and interaction of resources.

Intuitively, $A \otimes B$ describes the joint availability of two independent resources, while $A \parr B$ captures their dual form of interaction. Under a process interpretation, $\otimes$ can be read as producing a value of type A and continuing as B, whereas $A \parr B$ corresponds to receiving a value of type A and proceeding as B. The units 1 and $\bot$ represent the absence of further output and input, respectively. A simplified presentation of these rules is given below:

$$\frac{}{\vdash 1}\,1 \qquad \frac{\vdash \Gamma}{\vdash \Gamma, \bot}\,\bot \qquad \frac{\vdash \Gamma, A \quad \vdash \Delta, B}{\vdash \Gamma, \Delta, A \otimes B}\,\otimes \qquad \frac{\vdash \Gamma, A, B}{\vdash \Gamma, A \parr B}\,\parr$$

Here rule $\otimes$ presents a simplified form of the tensor rule from classical linear logic. In general, the tensor connective $A \otimes B$ combines two independent derivations. It requires a proof of A and a proof of B, each obtained from disjoint contexts, and produces a proof of the composite resource $A \otimes B$. Intuitively, this corresponds to combining two independent resources into a single composite resource, and can be read as producing a value of type A and then continuing as B.

The unit rule rule 1 introduces the unit 1, which represents the absence of further obligations. It can be understood as an "empty output", corresponding to a computation that performs no further interaction, and serves as the neutral element of tensor.

Dually, rules $\bot$ and $\parr$ describe the behavior of the connective $A \parr B$ and its unit $\bot$. The $\parr$ rule combines resources within a single context, reflecting a form of interaction where both components

are made available to the surrounding context. From an operational perspective, this corresponds to interacting with an input of type A while continuing as B. The unit $\bot$ represents the absence of further input, acting as an "empty input" and serving as the dual neutral element.

For example, two independent instances of the unit 1 can be introduced by the rule 1 and combined via rule $\otimes$ to derive $1 \otimes 1$:

$$\dfrac{\dfrac{}{\vdash 1}\,1 \qquad \dfrac{}{\vdash 1}\,1}{\vdash 1 \otimes 1}\,\otimes$$

Here each leaf introduces one resource independently, and the tensor rule combines them into a single compound resource $1 \otimes 1$. After this step, the individual resources are no longer available separately, reflecting the resource-sensitive nature of the system.

This resource-oriented view of logic provides the foundation for its applications in computer science, particularly in the study of concurrency and session-typed communication. In the following sections, this perspective is used to motivate the process interpretation underlying πLL.

2.2 The π-Calculus: Processes and Communication

The π-calculus [Milner et al. 1992] is a process calculus for modeling concurrent computation with dynamic communication topology, where channel names themselves can be transmitted, allowing the communication network to evolve during execution. We follow the presentation of Montesi and Peressotti [2021], which adopts Wadler's convention of using square brackets '[-]' for output and round parentheses '(-)' for input.

Programs in πMLL are processes (P, Q, R, ...), which communicate over names (x, y, z, ...) representing session endpoints. Sessions are binary (they involve exactly two endpoints), a discipline enforced by the typing system introduced in 2.3. We assume an infinite set of names with decidable equality. The process terms of πMLL are defined by the following grammar:

$P, Q ::=$	$x[y].P$	*output y on x and continue as P*
$\mid$	$x(y).P$	*input y on x and continue as P*
$\mid$	$x[].P$	*output (empty message) on x and continue as P*
$\mid$	$x().P$	*input (empty message) on x and continue as P*
$\mid$	$\boldsymbol{\nu} xy\, P$	*name restriction (cut)*
$\mid$	$P \mid Q$	*parallel composition*
$\mid$	$x[\mathrm{L}].P$	*select left on x and continue as P*
$\mid$	$x[\mathrm{R}].P$	*select right on x and continue as P*
$\mid$	$x.\mathrm{case}\{\mathrm{L}:P, \mathrm{R}:Q\}$	*offer a binary choice between P or Q on x*
$\mid$	$x[A].P$	*output type A on x and continue as P*
$\mid$	$x(X).P$	*input a type as X on x and continue as P*
$\mid$	$!x\langle z_1, \ldots, z_n\rangle.\{P\}$	*server (replicable process) on x depending on servers on $z_1 \ldots z_n$*
$\mid$	$x[\mathrm{USE}].P$	*consume a server on x and continue as P*
$\mid$	$x[\mathrm{DUP}](x').P$	*duplicate server x as x' in P*
$\mid$	$x[\mathrm{DISP}].P$	*dispose of a server on x*
$\mid$	$x \leftrightarrow y$	*link x and y*
$\mid$	$\mathbf{0}$	*terminated process*

Term $x[y].P$ allocates a fresh name y, outputs it over x, and proceeds as P. Dually, $x(y).P$ inputs a name over x, binding it to y in P. Terms $x[].P$ and $x().P$ model empty output and input. Restriction

$\boldsymbol{\nu} xy\,P$ connects the two endpoints x and y of P to form a session, where the order of x and y is irrelevant and $\boldsymbol{\nu} xy\,P = \boldsymbol{\nu} yx\,P$. Parallel composition $P \mid Q$ runs P and Q concurrently, and $\mathbf{0}$ is the terminated process, acting as the unit of parallel composition.

Example 2.1. Consider the process $P \triangleq \boldsymbol{\nu} xz\,(x[].\mathbf{0} \mid z().y[].\mathbf{0})$. This process creates a private session between the restricted endpoints x and z. The left process, $x[].\mathbf{0}$, signals termination on x before terminating, while the right process, $z().y[].\mathbf{0}$, waits for a termination signal on z before proceeding by sending a termination signal on y and then terminating.

This calculus provides the operational foundation for πLL. In the following section, we introduce the type system that governs how names are used, ensuring that each session endpoint is used exactly according to its protocol.

2.3 Types and Propositions-as-Sessions Correspondence

In πLL, session types are linear logic propositions [Caires and Pfenning 2010; Montesi and Peressotti 2021; Wadler 2014]. Each type describes the communication protocol of a channel endpoint, and duality, the key structural feature of classical linear logic, ensures that the two endpoints of every session implement complementary protocols.

The correspondence between linear logic and session types as presented in [Montesi and Peressotti 2021] following [Carbone et al. 2016; Wadler 2014] is captured directly by the type grammar. Each connective denotes a communication action, where $A \otimes B$ describes sending a value of type A and continuing as B, while its dual $A \parr B$ describes receiving. The units $\mathbf{1}$ and $\bot$, again, represent the absence of out and input, respectively. The additive connectives $A \oplus B$ and $A \,\&\, B$ capture choice in the sense of selection and offering of alternatives, while the exponentials $!A$ and $?A$ govern replication. The $\forall X.A$ and $\exists X.A$ allow polymorphic session behavior. The type grammar of πLL is defined as follows:

$A, B ::=$	$A \otimes B$	*send A, proceed as B*	$\mid$	$A \parr B$	*receive A, proceed as B*
$\mid$	$\mathbf{1}$	*empty output, unit for* $\otimes$	$\mid$	$\bot$	*empty input, unit for* $\parr$
$\mid$	$A \,\&\, B$	*offer A or B*	$\mid$	$A \oplus B$	*select A or B*
$\mid$	$\exists X.A$	*existential, type output*	$\mid$	$\forall X.A$	*universal, type input*
$\mid$	$?A$	*client request*	$\mid$	$!A$	*server accept*
$\mid$	X	*type variable*	$\mid$	${}^{\bot}X$	*dual of type variable*

Duality is defined structurally on session types by exchanging each connective with its dual and forms an involution, i.e. $(A^{\bot})^{\bot} = A$. Consequently, the endpoints of a well-typed session are always assigned dual types, ensuring they are compatibility:

$$(A \otimes B)^{\bot} = A^{\bot} \parr B^{\bot} \qquad \mathbf{1}^{\bot} = \bot \qquad (A \oplus B)^{\bot} = A^{\bot} \,\&\, B^{\bot}$$
$$(!A)^{\bot} = ?A^{\bot} \qquad (\exists X.A)^{\bot} = \forall X.A^{\bot} \qquad (X)^{\bot} = X^{\bot}$$

2.4 Environments, Hyperenvironments and Judgements

Typing in πLL is organised into two layers. Environments track how individual names are typed, while hyperenvironments track which names are used independently in parallel.

Following the presentation in [Montesi and Peressotti 2021], *environments* ($\Gamma, \Delta, \Xi, \ldots$) are defined as lists allowing exchange, meaning order is irrelevant. They can be written as $\Gamma = x_1 : A, \ldots, x_n : A_n$, where each x_i is associated to its respective A_i, for $1 \leq i \leq n$. Environments can be merged as long as they do not share any names for example assume Γ and Δ do not share any names, then Γ, Δ is the

environment containing exactly the associations in Γ and Δ regardless of ordering. Environments act as a partial commutative monoid, using "," as the sum and the empty environment $\bullet$ as unit:

$$\Gamma, \bullet = \Gamma \qquad \Gamma, \Delta = \Delta, \Gamma \qquad (\Gamma, \Delta), \Xi = \Gamma, (\Delta, \Xi)$$

Environments dictate how names are used, but it does not distinguish whether names are used independently or in parallel. That role is handled by handled by *hyperenvironments* ($\mathcal{G}, \mathcal{H}, \mathcal{I}, \ldots$), which are collections of environments joined using the parallel operator '$\mid$'. They can be written as follows $\mathcal{G} = \Gamma_1, \ldots, \Gamma_n$, where $\Gamma_1, \ldots, \Gamma_n$ is a collection of environments. Given $\mathcal{G}$ and $\mathcal{H}$ having no names in common then $\mathcal{G} \mid \mathcal{H}$ is the hyperenvironment containing exactly the environments of $\mathcal{G}$ and $\mathcal{H}$, ignoring order. Like environments, all names in a hyperenvironment are unique, they can be combined as long as they do not share any names, and they act as a partial commutative monoid using '$\mid$' the sum and $\varnothing$ as unit, where $\varnothing$ is the hyperenvironment containing no environments:

$$\mathcal{G} \mid \varnothing = \mathcal{G} \qquad \mathcal{G} \mid \mathcal{H} = \mathcal{H} \mid \mathcal{G} \qquad (\mathcal{G} \mid \mathcal{H}) \mid \mathcal{I} = \mathcal{G} \mid (\mathcal{H} \mid \mathcal{I})$$

Judgements, written as $\vdash P :: \mathcal{G}$, states that the process P uses its free names in accordance with $\mathcal{G}$ and can be derived using the inference rules displayed in Figure 1.

Typing Derivations ($\mathcal{D}, \mathcal{E}, \mathcal{F}, \ldots$) are written as $\overset{\mathcal{D}}{\vdash P :: \mathcal{G}}$ and are read as the derivation $\mathcal{D}$ having the judgement $\vdash P :: \mathcal{G}$ as its conclusion. The meaning of this statement is that the process, P, appearing in the judgement concluding a derivation is well-typed.

Multiplicative Fragment

$$\frac{\vdash P :: \mathcal{G} \mid \Gamma, x : A \mid \Delta, y : A^{\perp}}{\vdash \nu xy\, P :: \mathcal{G} \mid \Gamma, \Delta}\ \textsc{cut} \qquad \frac{\vdash P :: \mathcal{G} \quad \vdash Q :: \mathcal{H}}{\vdash P \mid Q :: \mathcal{G} \mid \mathcal{H}}\ \textsc{mix} \qquad \frac{}{\vdash \mathbf{0} :: \varnothing}\ \textsc{mix}_0$$

$$\frac{\vdash P :: \Gamma, y : A \mid \Delta, x : B}{\vdash x[y].P :: \Gamma, \Delta, x : A \otimes B}\ \otimes \qquad \frac{\vdash P :: \varnothing}{\vdash x[].P :: x : 1}\ 1 \qquad \frac{\vdash P :: \Gamma, y : A, x : B}{\vdash x(y).P :: \Gamma, x : A \parr B}\ \parr \qquad \frac{\vdash P :: \Gamma}{\vdash x().P :: \Gamma, x : \bot}\ \bot$$

Additional rules for Additives, Quantifiers and Exponentials

$$\frac{\vdash P :: \Gamma, x : A}{\vdash x[\textsc{l}].P :: \Gamma, x : A \oplus B}\ \oplus_1 \qquad \frac{\vdash P :: \Gamma, x : B}{\vdash x[\textsc{r}].P :: \Gamma, x : A \oplus B}\ \oplus_2 \qquad \frac{\vdash P :: \Gamma, x : A \quad \vdash Q :: \Gamma, x : B}{\vdash x.\mathrm{case}\{\textsc{l} : P, \textsc{r} : Q\} :: \Gamma, x : A \,\&\, B}\ \&$$

$$\frac{}{\vdash x \leftrightarrow y :: x : A^{\perp}, y : A}\ \textsc{ax} \qquad \frac{\vdash P :: \Gamma, x : A}{\vdash x[\textsc{use}].P :: \Gamma, x : ?A}\ ? \qquad \frac{\vdash P :: \Gamma}{\vdash x[\textsc{disp}].P :: \Gamma, x : ?A}\ \textsc{w} \qquad \frac{\vdash P :: \Gamma, x : ?A, x' : ?A}{\vdash x[\textsc{dup}](x').P :: \Gamma, x : ?A}\ \textsc{c}$$

$$\frac{\vdash P :: z_1 : ?B_1, \ldots, z_n : ?B_n, x : A}{\vdash !x\langle z_1, \ldots, z_n \rangle.\{P\} :: z_1 : ?B_1, \ldots, z_n : ?B_n, x : !A}\ ! \qquad \frac{\vdash P :: \Gamma, x : B\{A/X\}}{\vdash x[A].P :: \Gamma, x : \exists X.B}\ \exists \qquad \frac{\vdash P :: \Gamma, x : B \quad X \notin \mathrm{ft}(\Gamma)}{\vdash x(X).P :: \Gamma, x : \forall X.B}\ \forall$$

Fig. 1. πLL, typing rules.

The multiplicative rules form the core of the system. Rule CUT composes two processes in P sharing dual endpoints $x : A$ and $y : A^{\perp}$, hiding them via restriction. This is the logical equivalent of communication. Rule MIX and rule $\textsc{mix}_0$ compose independent processes in parallel and introduce the terminated process, respectively.

Rule $\otimes$ and rule $\parr$ type output and input. Note that rule $\otimes$ types the sent name y in a *separate* environment from the continuation, reflecting that the two are independent resources. By contrast, rule $\parr$ keeps y and the continuation x in the *same* environment, reflecting their sequential

dependency. The units rule 1 and rule $\bot$ type empty output and empty input, where rule 1 requires the continuation to be well-typed in the empty hyperenvironment, meaning P is necessarily semantically equivalent to $\mathbf{0}$.

The additive rules (rule $\oplus_1$, rule $\oplus_2$, and rule $\&$) type selection and offering of alternative processing paths. Note that rule $\&$ requires both branches to be typed in the *same* context Γ, reflecting that the choice of branch is made by the other endpoint.

The exponential rules type server replication. The rule ! requires all free names except x to be typed using ?, enforcing that a server only depends on other servers. The rule ?, rule w, and rule c allow a client to use, discard, or duplicate a server respectively.

The rule AX types a link between two dual endpoints, $x{\leftrightarrow}y$, forwarding all messages sent on x safely to y and vice versa. Rule $\exists$ and rule $\forall$ type polymorphic communication, where the side condition $X \notin \mathrm{ft}(\Gamma)$ in rule $\forall$, ensures the introduced type variable, X, does not accendentally shadow one already existing in Γ.

Example 2.2. The process from Example 2.1 has the typing $\vdash \boldsymbol{\nu} zx\,(z().y[].\mathbf{0} \mid x[].\mathbf{0}) :: y : 1$, as shown by the derivation below:

$$\dfrac{\dfrac{\dfrac{}{\vdash \mathbf{0} :: \varnothing}\,\mathrm{MIX}_0}{\vdash x[].\mathbf{0} :: x : 1}\,1 \qquad \dfrac{\dfrac{\dfrac{}{\vdash \mathbf{0} :: \varnothing}\,\mathrm{MIX}_0}{\vdash y[].\mathbf{0} :: y : 1}\,1}{\vdash z().y[].\mathbf{0} :: y : 1, z : \bot}\,\bot}{\vdash x[].\mathbf{0} \mid z().y[].\mathbf{0} :: x : 1 \mid y : 1, z : \bot}\,\mathrm{MIX}}{\vdash \boldsymbol{\nu} xz\,(x[].\mathbf{0} \mid z().y[].\mathbf{0}) :: y : 1}\,\mathrm{CUT}$$

Note that, for the typing context $x : 1 \mid y : 1, z : \bot$ to align with the premise of rule CUT, we utilise the monoidal properties of environments and hyperenvironments. In particular, using their identity laws, we instantiate the meta-variables of the rule as $\mathcal{G} = \varnothing$ and $\Gamma = \bullet$. Under these instantiations, the premise of rule CUT: $\vdash P :: \mathcal{G} \mid \Gamma, x : A \mid \Delta, y : A^{\perp}$ can be viewed as $x : A \mid \Delta, y : A^{\perp}$. This effectively instantiates the CUT over the dual types $A = 1$ and $A^{\perp} = \bot$ on channels x and y, with $\Gamma = y : 1$, while ensuring the restricted endpoints implement dual session behaviours, fulfilling the requirements for a creating a valid CUT (session).

2.5 Operational Semantics

The operational semantics of πLL are given as a labelled transition system (LTS) defined over *typing derivations*, in which processes evolve by performing actions on their free names. The semantics follow a Structural Operational Semantics (SOS) style presentation and form the basis for the metatheoretic results of the calculus. We will be focusing on the multiplicative fragment for this part, which is given in Figure 2. The full LTS is given in Montesi and Peressotti [2021].

Remark 2.3. The operational semantics rely explicitly on the structural equivalence (rule $=_\alpha$) to rename restricted variables via α-equivalence prior to a transition. In paper-based presentations, α-equivalence is often handled implicitly through Barendregt's Variable Convention. In a mechanised setting, however, reasoning over named variables requires freshness conditions, capture-avoiding substitution, and explicit renaming lemmas, introducing substantial proof overhead [Aydemir et al. 2005; Charguéraud 2012]. As discussed in Section 3, this motivates the use of a different binding representation for the syntax of πLL, one designed to reduce explicit α-equivalence reasoning and simplify the treatment of bound variables.

Actions are defined in the set $Act = \{x[], x(), x[y], x(y) \mid x, y \text{ names}\}$, representing empty output, empty input, name output, and name input respectively. These come together with the silent label τ for internal communication and the parallel composition of labels $l \mid l'$ to form the

$$\frac{\mathcal{D}}{\vdash P :: \varnothing}{\vdash x[].P :: x : 1}\,1 \xrightarrow{x[]} \mathcal{D} \vdash P :: \varnothing \qquad \frac{\mathcal{D}}{\vdash P :: \Gamma, x' : A \mid \Delta, x : B}{\vdash x[x'].P :: \Gamma, \Delta, x : A \otimes B}\,\otimes \xrightarrow{x[x']} \mathcal{D} \vdash P :: \Gamma, x' : A \mid \Delta, x : B$$

$$\frac{\mathcal{D}}{\vdash P :: \Gamma}{\vdash x().P :: \Gamma, x : \bot}\,\bot \xrightarrow{x()} \mathcal{D} \vdash P :: \Gamma \qquad \frac{\mathcal{D}}{\vdash P :: \Gamma, x' : A, x : B}{\vdash x(x').P :: \Gamma, x : A \parr B}\,\parr \xrightarrow{x(x')} \mathcal{D} \vdash P :: \Gamma, x' : A, x : B$$

$$\frac{\mathcal{D} \vdash P :: \mathcal{G} \xrightarrow{l} \mathcal{D}' \vdash P' :: \mathcal{G}' \qquad \mathrm{i}(l) \cap \mathrm{f}(Q) = \varnothing}{\dfrac{\mathcal{D} \vdash P :: \mathcal{G} \quad \mathcal{E} \vdash Q :: \mathcal{H}}{\vdash P \mid Q :: \mathcal{G} \mid \mathcal{H}}\,\mathrm{MIX} \xrightarrow{l} \dfrac{\mathcal{D}' \vdash P' :: \mathcal{G}' \quad \mathcal{E} \vdash Q :: \mathcal{H}}{\vdash P' \mid Q :: \mathcal{G}' \mid \mathcal{H}}\,\mathrm{MIX}}\,\mathrm{PAR}_1$$

$$\frac{\mathcal{E} \vdash Q :: \mathcal{H} \xrightarrow{l} \mathcal{E}' \vdash Q' :: \mathcal{H}' \qquad \mathrm{i}(l) \cap \mathrm{f}(P) = \varnothing}{\dfrac{\mathcal{D} \vdash P :: \mathcal{G} \quad \mathcal{E} \vdash Q :: \mathcal{H}}{\vdash P \mid Q :: \mathcal{G} \mid \mathcal{H}}\,\mathrm{MIX} \xrightarrow{l} \dfrac{\mathcal{D} \vdash P :: \mathcal{G} \quad \mathcal{E}' \vdash Q' :: \mathcal{H}'}{\vdash P \mid Q' :: \mathcal{G} \mid \mathcal{H}'}\,\mathrm{MIX}}\,\mathrm{PAR}_2$$

$$\frac{\mathcal{D} \vdash P :: \mathcal{G} \xrightarrow{l} \mathcal{D}' \vdash P' :: \mathcal{G}' \qquad \mathcal{E} \vdash Q :: \mathcal{H} \xrightarrow{l'} \mathcal{E}' \vdash Q' :: \mathcal{H}' \qquad \mathrm{i}(l|l') \cap \mathrm{f}(P \mid Q) = \varnothing}{\dfrac{\mathcal{D} \vdash P :: \mathcal{G} \quad \mathcal{E} \vdash Q :: \mathcal{H}}{\vdash P \mid Q :: \mathcal{G} \mid \mathcal{H}}\,\mathrm{MIX} \xrightarrow{l|l'} \dfrac{\mathcal{D}' \vdash P' :: \mathcal{G}' \quad \mathcal{E}' \vdash Q' :: \mathcal{H}'}{\vdash P' \mid Q' :: \mathcal{G}' \mid \mathcal{H}'}\,\mathrm{MIX}}\,\mathrm{SYN}$$

$$\frac{P =_\alpha Q \qquad \mathcal{E} \vdash Q :: \mathcal{G} \xrightarrow{l} \mathcal{E}' \vdash Q' :: \mathcal{G}'}{\mathcal{D} \vdash P :: \mathcal{G} \xrightarrow{l} \mathcal{E}' \vdash Q' :: \mathcal{G}'}\,=_\alpha \qquad \frac{\dfrac{\mathcal{D} \vdash P :: \mathcal{G} \mid x : 1 \mid \Gamma, y : \bot \xrightarrow{x[]\mid y()} \mathcal{D}' \vdash P' :: \mathcal{G} \mid \Gamma}{\vdash P :: \mathcal{G} \mid x : 1 \mid \Gamma, y : \bot}\,1\bot}{\vdash \nu xy\, P :: \mathcal{G} \mid \Gamma}\,\mathrm{CUT} \xrightarrow{\tau} \mathcal{D}' \vdash P' :: \mathcal{G} \mid \Gamma$$

$$\frac{\mathcal{D} \vdash P :: \mathcal{G} \mid \Gamma, \Delta, x : A \otimes B \mid \Xi, y : A^\bot \parr B^\bot \xrightarrow{x[x']\mid y(y')} \mathcal{D}' \vdash P' :: \mathcal{G} \mid \Gamma, x : B \mid \Delta, x' : A \mid \Xi, y : B^\bot, y' : A^\bot}{\dfrac{\mathcal{D}}{\vdash P :: \mathcal{G} \mid \Gamma, \Delta, x : A \otimes B \mid \Xi, y : A^\bot \parr B^\bot}\,\mathrm{CUT} \xrightarrow{\tau} \dfrac{\mathcal{D}'}{\dfrac{\vdash P' :: \mathcal{G} \mid \Gamma, x : B \mid \Delta, x' : A \mid \Xi, y : B^\bot, y' : A^\bot}{\vdash \nu x' y'\, P' :: \mathcal{G} \mid \Gamma, x : B \mid \Delta, \Xi, y : B^\bot}\,\mathrm{CUT}}{\vdash \nu xy\, \nu x' y'\, P' :: \mathcal{G} \mid \Gamma, \Delta, \Xi}\,\mathrm{CUT}}\,\otimes\parr$$

$$\frac{\mathcal{D} \vdash P :: \mathcal{G} \mid \Gamma, x : A \mid \Delta, y : A^\bot \xrightarrow{l} \mathcal{D}' \vdash P' :: \mathcal{G}' \mid \Gamma', x : A \mid \Delta', y : A^\bot \qquad x, y \notin \mathrm{f}(l) \cup \mathrm{i}(l)}{\dfrac{\mathcal{D}}{\vdash P :: \mathcal{G} \mid \Gamma, x : A \mid \Delta, y : A^\bot}\,\mathrm{CUT} \xrightarrow{l} \dfrac{\mathcal{D}'}{\vdash P' :: \mathcal{G}' \mid \Gamma', x : A \mid \Delta', y : A^\bot}\,\mathrm{CUT}}\,\mathrm{RES}$$

Fig. 2. πMLL, transition rules for derivations.

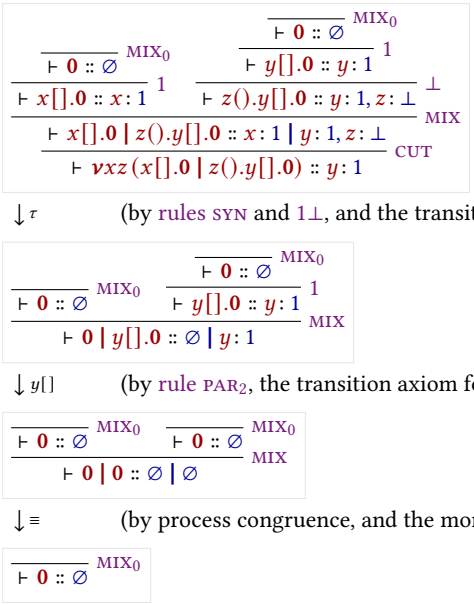

Fig. 3. An execution for the derivation in Example 2.2.

set of transition labels, defined as $Lbl = \{\tau, l, (l|l') \mid l, l' \in Act, \mathrm{i}(l) \cap \mathrm{i}(l') = \emptyset\}$. The restriction imposed on the parallel composition of labels exists to ensure two labels do not introduce the same name, which would break linear resource management.

Example 2.4. Figure 3 shows a possible execution for the derivation of the process from Example 2.1.

Each logical rule (1, $\bot$, $\otimes$, $\parr$) has a corresponding transition axiom. These follow the prefix-continuation pattern $\pi.P \xrightarrow{\pi} P$, where performing the action π leads to the remainder of the process. Rule SYN allows for the pairing of concurrent actions via $l|l'$, which is used to facilitate communication by letting two endpoints connected by a restriction perform compatible actions. When two compatible actions are performed over endpoints connected by a restriction, rules 1$\bot$ and $\otimes\parr$ simplify the term into an internal τ-transition, modeling the synchronisation of the session.

2.6 Process Semantics and Erasure

While the typing derivations of πLL dictate the valid interactions within a session, the underlying process terms possess a standard, untyped operational behavior that closely mirrors the classic π-calculus. To formally relate the typed derivations to their untyped execution, we follow [Montesi and Peressotti 2021] and define an erasure function, $proc : Der \to Proc$, which strips away the typing information from a derivation. Recursively, $proc$ maps the head of a typing derivation to the specific process constructor it introduces, continuing structurally onto the process's continuation. For any well-typed πLL term, $proc(\mathcal{D})$ effectively extracts the pure process term from the derivation's conclusion.

Example 2.5. Consider the well-typed process P formed by adding a third parallel component to the derivation of Example 2.2 before the cut (A derivation of which can be seen on Figure 4):

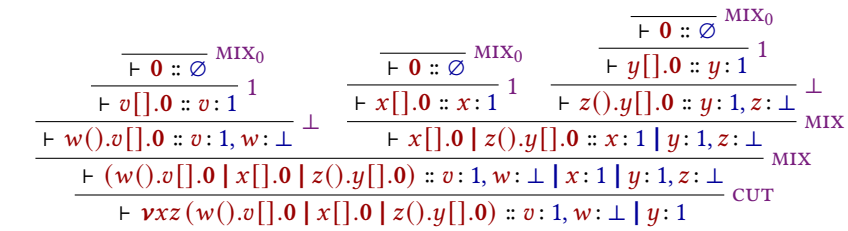

Fig. 4. A derivation for the process $\nu xz\, w().y[].\mathbf{0} \mid x[].\mathbf{0} \mid z().y[].\mathbf{0}$

$$P = \nu xz\,(w().v[].\mathbf{0} \mid x[].\mathbf{0} \mid z().y[].\mathbf{0})$$

The component containing w and v is independent from the component containing x, z, and y. This allows their respective actions to interleave freely, resulting in the following execution traces, illustrating the concurrency of the calculus:

$$P \xrightarrow{\tau} \xrightarrow{w()} \xrightarrow{v[]} \xrightarrow{y[]} \mathbf{0} \qquad P \xrightarrow{\tau} \xrightarrow{w()} \xrightarrow{y[]} \xrightarrow{v[]} \mathbf{0} \qquad P \xrightarrow{\tau} \xrightarrow{y[]} \xrightarrow{w()} \xrightarrow{v[]} \mathbf{0}$$

$$P \xrightarrow{w()} \xrightarrow{v[]} \xrightarrow{\tau} \xrightarrow{y[]} \mathbf{0} \qquad P \xrightarrow{w()} \xrightarrow{\tau} \xrightarrow{v[]} \xrightarrow{y[]} \mathbf{0} \qquad P \xrightarrow{w()} \xrightarrow{\tau} \xrightarrow{y[]} \xrightarrow{v[]} \mathbf{0}$$

The semantics for process terms must inherently agree with the semantics of derivations for well-typed terms. As formulated in [Montesi and Peressotti 2021], this relationship is established functorially. Let $\mathcal{P}(X) = \{X' \mid X' \subseteq X\}$ and $\mathcal{L}(X) = X \times Lbl$ be endofunctors representing the states and labeled transitions, respectively. The erasure function *proc* acts as a homomorphism from LTS of derivations (lts_d) to the LTS of processes (lts_p), ensuring that the square of these LTS mappings commutes.

In the context of operational semantics, this functorial homomorphism yields an Erasure theorem. It guarantees that the Labeled Transition System for derivations and the SOS for untyped processes simulate each other exactly:

Theorem 2.6 (Erasure). *The process LTS (lts_p) enjoys erasure with respect to the derivation LTS (lts_d). Concretely, for any valid typing derivation $\mathcal{D}$ and label $l \in Lbl$:*

(1) ***Soundness of Erasure:*** *If $\mathcal{D} \xrightarrow{l} \mathcal{D}'$, then $proc(\mathcal{D}) \xrightarrow{l} proc(\mathcal{D}')$.*

(2) ***Completeness of Erasure:*** *If $proc(\mathcal{D}) \xrightarrow{l} P'$, then there exists a derivation $\mathcal{D}'$ such that $\mathcal{D} \xrightarrow{l} \mathcal{D}'$ and $proc(\mathcal{D}') = P'$.*

Corollary 2.7 (Typability Preservation). *If P is well-typed and $P \xrightarrow{l} P'$, then P' is well-typed.*

This theorem allows us to define the operational target for πLL processes without the syntactic overhead of typing environments. Using the erasure formulation, we can transform the operational semantics for derivations directly into an SOS specification for processes.

Remark 2.8. The untyped LTS for processes relies heavily on explicit side conditions regarding free and introduced names to prevent the collision of bound variables (e.g., ensuring $i(l) \cap i(l') = \emptyset$). However, when operating at the level of typing derivations, these side conditions become inherently redundant. Because πLL enforces linear resource management, the structural rules of the typed LTS dictate that hyperenvironments can only be merged when their domains are strictly disjoint. Consequently, any well-typed derivation inherently satisfies the name-distinctness and freshness

$$x[x'].P \xrightarrow{x[x']} P \qquad x(x').P \xrightarrow{x(x')} P \qquad x[].P \xrightarrow{x[]} P \qquad x().P \xrightarrow{x()} P$$

$$\frac{P \xrightarrow{l} P' \quad \mathrm{i}(l) \cap \mathrm{f}(Q) = \emptyset}{P \mid Q \xrightarrow{l} P' \mid Q}\ \mathrm{PAR}_1 \qquad \frac{Q \xrightarrow{l} Q' \quad \mathrm{i}(l) \cap \mathrm{f}(P) = \emptyset}{P \mid Q \xrightarrow{l} P \mid Q'}\ \mathrm{PAR}_2$$

$$\frac{P \xrightarrow{l} P' \quad Q \xrightarrow{l'} Q' \quad \mathrm{i}(l|l') \cap \mathrm{f}(P \mid Q) = \emptyset}{P \mid Q \xrightarrow{l|l'} P' \mid Q'}\ \mathrm{SYN} \qquad \frac{P \xrightarrow{x[x']|y(y')} P'}{\nu xy P \xrightarrow{\tau} \nu xy\, \nu x'y'\, P'}\ \otimes\!\!\parr$$

$$\frac{P \xrightarrow{x[]|y()} P'}{\nu xy P \xrightarrow{\tau} P'}\ 1\bot \qquad \frac{P \xrightarrow{l} P' \quad x, y \notin \mathrm{f}(l) \cup \mathrm{i}(l)}{\nu xy P \xrightarrow{l} \nu xy P'}\ \mathrm{RES} \qquad \frac{P =_\alpha Q \quad Q \xrightarrow{l} Q'}{P \xrightarrow{l} Q'}\ =_\alpha$$

$$\mathrm{i}(l|l') = \mathrm{i}(l) \cup \mathrm{i}(l') \qquad \mathrm{f}(l|l') = \mathrm{f}(l) \cup \mathrm{f}(l')$$
$$\mathrm{i}(\tau) = \emptyset \qquad \mathrm{f}(\tau) = \emptyset$$
$$\mathrm{i}(x[]) = \mathrm{i}(x()) = \emptyset \qquad \mathrm{f}(x[]) = \mathrm{f}(x()) = \{x\}$$
$$\mathrm{i}(x[x']) = \mathrm{i}(x(x')) = \{x'\} \qquad \mathrm{f}(x[x']) = \mathrm{f}(x(x')) = \{x\}$$

Fig. 5. πMLL, transition rules for processes.

requirements of the underlying untyped calculus, shifting the burden of safety from dynamic transition checks to static structural typing.

Proposition 2.9. *The operational semantics of πLL processes strictly respects the introduction of names via transition labels. Formally, if $P \xrightarrow{l} P'$, then $\mathrm{i}(l) \cap \mathrm{f}(P) = \emptyset$, and If P is well-typed, then $\mathrm{i}(l) = \mathrm{f}(P') \setminus \mathrm{f}(P)$ (TODO: Evaluate relevance - delete depending on if this property is mentioend in later proofs).*

Remark 2.10. Proposition 2.9 is vital for mechanising the metatheory of πLL. Because proof assistants tools require explicit tracking of fresh and bound variable to prevent accidental capture, this proposition acts as the formal bridge between the dynamic semantics and the static typing. It guarantees that a well-typed process evolves predictably, ensuring that it does not spontaneously generate resources during a transition. (TODO: same as Proposition 2.9)

2.7 Semantics of Typing Environments

Just as processes evolve through computation, the typing environments that govern them must also evolve to accurately track the state of a session. In πLL, types capture communication protocols. Consequently, the transition rules for typing environments specify exactly which actions are permitted and how the available resources are consumed by those actions.

Similar to how we defined the process erasure function, we again follow Montesi and Peressotti [2021] and define a function $env : Der \to Env$ that maps a valid typing derivation directly to the hyperenvironment in its conclusion (i.e., $env(\vdash P :: \mathcal{G}) = \mathcal{G}$). Using this extraction, Montesi and Peressotti [2021] derive a LTS specifically for typing environments (lts_e), shown in Figure 6.

In this LTS, observable actions actively consume resources from the environment. For example, if a hyperenvironment contains a waiting channel $x : \bot$, performing an input action $x()$ will consume this type, removing it from the resulting environment. Conversely, internal communications do not alter the available external resources. This is formalised by the τ-reflexivity axiom, $\mathcal{G} \mid \Gamma \xrightarrow{\tau} \mathcal{G} \mid \Gamma$, which states that any valid hyperenvironment transitions to itself under a silent τ-action.

The requirement that a process can only perform actions permitted by its typing environment is known as **Session Fidelity**. Just as the *proc* function established operational correspondence for processes, the *env* function establishes a correspondence for environments.

While the rules in Figure 6 define how environments can transition in isolation, the execution of a well-typed program requires the process and its environment to step in unison. This co-dependent relationship has processes only perform actions permitted by its typing environment, while the

$$x:1 \xrightarrow{x[]} \varnothing \qquad \Gamma, x:\bot \xrightarrow{x()} \Gamma \qquad \Gamma, \Delta, x:A \otimes B \xrightarrow{x[x']} \Gamma, x':A \mid \Delta, x:B \qquad \Gamma, x:A \parr B \xrightarrow{x(x')} \Gamma, x':A, x:B$$

$$\frac{\mathcal{G} \xrightarrow{l} \mathcal{G}'}{\mathcal{G} \mid \mathcal{H} \xrightarrow{l} \mathcal{G}' \mid \mathcal{H}}\ \text{PAR}_1 \qquad \frac{\mathcal{H} \xrightarrow{l} \mathcal{H}'}{\mathcal{G} \mid \mathcal{H} \xrightarrow{l} \mathcal{G} \mid \mathcal{H}'}\ \text{PAR}_2 \qquad \frac{\mathcal{G} \xrightarrow{l} \mathcal{G}' \quad \mathcal{H} \xrightarrow{l'} \mathcal{H}'}{\mathcal{G} \mid \mathcal{H} \xrightarrow{l|l'} \mathcal{G}' \mid \mathcal{H}'}\ \text{SYN}$$

$$\frac{\mathcal{G} \mid \Gamma, x:A^\bot \mid \Delta, y:A \xrightarrow{l} \mathcal{G}' \mid \Gamma', x:A^\bot \mid \Delta', y:A}{\mathcal{G} \mid \Gamma, \Delta \xrightarrow{l} \mathcal{G}' \mid \Gamma', \Delta'}\ \text{RES} \qquad \frac{\mathcal{G} \mid x:1 \mid \Gamma, y:\bot \xrightarrow{x[]|y()} \mathcal{G} \mid \Gamma}{\mathcal{G} \mid \Gamma \xrightarrow{\tau} \mathcal{G} \mid \Gamma}\ 1\bot$$

$$\frac{\mathcal{G} \mid \Gamma, \Delta, x:A \otimes B \mid \Xi, y:A^\bot \parr B^\bot \xrightarrow{x[x']|y(y')} \mathcal{G} \mid \Gamma, x:B \mid \Delta, x':A \mid \Xi, y:B^\bot, y':A^\bot}{\mathcal{G} \mid \Gamma, \Delta, \Xi \xrightarrow{\tau} \mathcal{G} \mid \Gamma, \Delta, \Xi}\ \otimes\parr$$

Fig. 6. πMLL, transition rules for typing environments.

environment dynamically mirrors the physical control flow of the process, and is formally captured by the property of **Session Fidelity**. Just as the *proc* function established operational correspondence for the untyped calculus, the *env* function establishes this exact behavioral correspondence for the typing contexts.

Theorem 2.11 (Session Fidelity). *Let $\mathcal{D}$ be a valid typing derivation such that $env(\mathcal{D}) = \mathcal{G}$ and $proc(\mathcal{D}) = P$. If the process transitions such that $P \xrightarrow{l} P'$, then the environment can mimic this transition*
$\mathcal{G} \xrightarrow{l} \mathcal{G}'$,
and there exists a new valid derivation $\mathcal{D}'$ for P' under $\mathcal{G}'$.

To illustrate Session Fidelity in action, we can observe how the environment is constrained to follow the exact execution traces of the process it types.

Example 2.12 (Session Fidelity in Execution). Recall the process derivation from Example 2.5, typed by the environment $\mathcal{G} = v:1, w:\bot \mid y:1$. The process begins by performing an observable input action on w. By Session Fidelity, the environment must faithfully mimic this action, consuming the corresponding type:

$$v:1, w:\bot \mid y:1 \xrightarrow{w()} v:1 \mid y:1$$

The resulting derivative of the process is now typed by this new environment. At this intermediate stage, if we were to view the environment LTS in isolation, the pure semantics would theoretically allow an output on y to fire immediately, as $y:1$ is an available, unblocked resource.

However, Session Fidelity dictates a strict co-dependence. The environment merely mirrors the process, and the underlying process term strictly guards the $y[]$ action behind a prefix, requiring a prior τ-synchronisation. Therefore, the environment must wait. When the process finally resolves its internal communication, the environment mimics it via the τ-reflexivity axiom, remaining structurally unchanged:

$$v:1 \mid y:1 \xrightarrow{\tau} v:1 \mid y:1$$

Only after this synchronisation unblocks the continuation can the process perform the $y[]$ action, which the environment once again mirrors by consuming the final resource:

$$v:1 \mid y:1 \xrightarrow{y[]} v:1$$

Remark 2.13 (The Mechanisation Gap). Session Fidelity is the ultimate touchstone for validating the safety of a session-typed language. However, verifying this property in a proof assistant exposes

a stark gap between paper mathematics and mechanised logic. On paper, the invariant that an environment *dynamically mirrors* a process is highly intuitive, and transitions are easily massaged using structural equivalences and implicit variable conventions. In a machine, however, formally proving that types are consumed correctly requires a rigorous, algorithmic representation of bound variables and explicit freshness. The theoretical elegance of the environment LTS cannot be mechanised without first establishing a concrete architectural foundation for these variables, which dictates the design choices discussed in the following chapters.

2.8 Managing Bound Variables

Having established the operational semantics of πLL and the ultimate goal of verifying Session Fidelity, we must address the treatment of bound variables and freshness, a central technical challenge that underpins the entire formalisation.

As we have seen, many constructs in πLL introduce scoped identifiers. Restriction binds pairs of session endpoints, input constructs bind received channel names, polymorphic communication binds type variables, and duplication constructs introduce fresh channel endpoints. These bindings appear throughout both the process syntax and the labelled transition system.

The concrete choice of bound and introduced names is semantically irrelevant provided the binding structure is preserved. For example, renaming a restricted channel does not change the behaviour of a process. However, reasoning formally about substitutions, transitions, and typing derivations requires these names, scopes, and freshness conditions to be tracked explicitly.

This principle is formalised as *α-equivalence*, which identifies terms that differ only in the names of their bound variables. For instance, the following two processes are α-equivalent:

$$\nu ab\,(a[].\mathbf{0} \mid b().x[].\mathbf{0}) =_\alpha \nu zw\,(z[].\mathbf{0} \mid w().x[].\mathbf{0})$$

The operational semantics of πLL relies on α-equivalence using rule $=_\alpha$ to rename bound variables on the fly, ensuring that restricted scopes and input prefixes do not accidentally capture free names as processes evolve.

Substitution raises this exact issue. Consider the substitution where the free name x is replaced by z in the right-hand process from the previous example:

$$\nu zw\,(z[].\mathbf{0} \mid w().x[].\mathbf{0})$$

Here z is already bound by the restriction, so naively replacing x with z would cause the restriction to capture the newly inserted z. Before the substitution can proceed, and because restriction binds session endpoints as pairs, both the bound z and w must be α-renamed to fresh names, such as to a and b. Similarly, any binder within the process that would capture the substituted name must also be renamed before substitution continues.

In paper-based presentations this is commonly handled by *Barendregt's variable convention*, which assumes bound variables are always chosen distinct from free variables and one another whenever convenient. This makes proofs readable but remains an informal meta-level assumption.

A proof assistant requires every freshness condition and renaming step to be represented and verified explicitly [Aydemir et al. 2005]. A direct mechanisation using named variables therefore introduces substantial proof overhead for capture avoidance and α-equivalence reasoning. To avoid this, the formalisation adopts a *locally nameless* representation of binding, which eliminates α-equivalence as a proof obligation by construction. This is the subject of the following chapter.

3 Binding Architecture

This chapter develops the locally nameless binding architecture used throughout the formalisation. We begin by stating the design goals that motivate the approach (Section 3.1), then introduce the

locally nameless representation and its core operations (Sections 3.2 and 3.3), and conclude with co-finite quantification (Section 3.4), which governs how inductive definitions interact with binders.

3.1 Design Goals and the Approach

This chapter introduces the binding representation used throughout the formalisation. The representation is motivated by three primary design goals:

(1) **Eliminate α-equivalence overhead.** α-equivalent terms should be represented identically, reducing α-equivalence reasoning to definitional equality within the proof assistant.
(2) **Preserve human readability.** Terms should remain explicit and concrete, keeping substitution and freshness reasoning close to their paper-based presentation.
(3) **Avoid informal meta-level conventions.** The representation should not rely on Barendregt's convention or unverified freshness assumptions; capture avoidance and freshness conditions should instead be enforced by construction.

Although both channels and type variables involve binding, they exhibit fundamentally different operational behaviour. Session channels participate in dynamic communication and substitution across concurrent processes, while polymorphic type variables remain statically scoped within typing derivations. The formalisation therefore adopts different binding strategies for these two classes of identifiers while maintaining a common underlying binding architecture.

To achieve these goals, we adopt a locally nameless binding architecture for the calculus [Charguéraud 2012], separating free identifiers from bound occurrences. Bound variables are represented using indices, causing α-equivalent terms to coincide structurally, while free variables remain explicit and readable.

This separation simplifies substitution by restricting it to free variables while leaving bound indices untouched, thereby avoiding accidental capture by construction. As a result, many proofs that traditionally require substantial renaming infrastructure reduce instead to structural recursion and index manipulations.

The remainder of this chapter develops the theoretical machinery underlying this representation.

3.2 The Locally Nameless Representation

The locally nameless representation, as presented by Charguéraud [2012], separates variables into two disjoint syntactic categories: free names and bound indices. This combines the readability of named syntax for free variables with a structural representation of binding.

(1) **Bound variables** are represented using *De Bruijn indices* [de Bruijn 1972], where each index denotes the distance to its corresponding binder. Since the representation is purely positional, α-equivalent terms become structurally identical.
(2) **Free variables** retain explicit names, preserving the readability of open terms and allowing standard name-based reasoning about environments and transition labels.

By structurally separating these two concepts, the locally nameless approach simplifies standard operations such as capture-avoiding substitution. Since free variables and bound variables inhabit distinct syntactic categories, substitution on free variables is defined to not interact with bound indices. We apply this theory to both the structural and operational layers of πLL.

LN Syntax for Types. In the original presentation of πLL, polymorphic types rely on standard named variables. Under the locally nameless architecture, we expand the syntax of type variables to explicitly distinguish between free and bound occurrences. The grammar for types is updated as

follows:

$$\begin{array}{rcll} T & ::= & \mathrm{free}(x) \mid \mathrm{bound}(k) & (\text{where } x, k \in \mathbb{N}) \\ A, B & ::= & \cdots \mid \exists.A \mid \forall.A \mid X \mid X^{\perp} & (\text{where } X \in T) \end{array}$$

Here, a free type variable carries a unique identifier x, while a bound type variable carries a De Bruijn index k. For example, the polymorphic type $\forall X.\exists Y.(X \parr Y)$ is structurally represented without names using the new syntax as $\forall.\exists.(\mathrm{bound}(1) \parr \mathrm{bound}(0))$, where 0 points to the inner existential binder and 1 points to the outer universal binder.

LN Syntax for Processes. Similarly, the standard process syntax displayed in Section 2.2 is updated to reflect the locally nameless separation of channel endpoints. A channel is split into its bound and free counterparts, where bound channel names are represented by indices, while free channels remain explicit identifiers.

Naively removing the bound names from the process syntax would turn binding operations such as $x[y].P$ and $x(y).P$ into $x[].P$ and $x().P$ making them syntactically indistinguishable from their non-binding unit counterparts. To combat this we annotate anonymous binders using placeholder symbols. We use $\bullet$ for a bound process name, and $\diamond$ for a bound type variable.

$$\begin{array}{rcll} \mathit{Channel} & ::= & \mathrm{free}(x) \mid \mathrm{bound}(k) & (\text{where } x, k \in \mathbb{N}) \\ P, Q & ::= & \cdots \mid x[\bullet].P \mid x(\bullet).P \mid \nu(\bullet,\bullet).\,P \mid x(\diamond).P & (\text{where } x \in \mathit{Channel}) \end{array}$$

While the visual anonymity of these constructors is evident from the grammar, their mechanical implementation relies on precise index management. For a multi-name binder like restriction, $\boldsymbol{\nu}(\bullet,\bullet)\,P$, the two dual endpoints are simultaneously mapped to distinct indices within the continuation of P, namely 0 and 1. For communication prefixes like $x(\bullet).P$, the payload is implicitly bound as index 0 within P, while the explicit free channel x is deliberately preserved as the subject. By maintaining explicit free names strictly for these outward-facing communications, the transition labels and operational semantics of the calculus remain completely unaffected by the architectural shift.

While the locally nameless architecture elegantly resolves α-equivalence, moving between abstracted and instantiated forms requires specialised operations. To execute a substitution or evaluate a process across a binder, we must formally define how to open and close these terms.

3.3 Opening and Closing Binders

Because the locally nameless representation separates free and bound variables, standard substitution cannot be applied uniformly under binders. Instead, moving between bound and free representations is handled by two dual operations, namely *opening* and *closing* [Charguéraud 2012].

Opening (Instantiation). Opening is the operation of traversing into a binding construct and replacing the exposed bound index with a concrete free name. Formally, opening a process P at depth k with a free name x is denoted $\{k \mapsto x\}P$.

The operation proceeds via structural recursion. A depth counter, initially set to 0, tracks the number of binders traversed. As the recursion descends through the syntax tree, the counter increments when passing a new binder. Consider being at depth k, then traversing a single-name binder like $x(\bullet).P$ increments the depth to $k+1$, while traversing a double-name binder like $\boldsymbol{\nu}(\bullet,\bullet)\,P$ would increment it to $k+2$.

When the recursion reaches a bound variable $\mathrm{bound}(n)$, the index n is compared against the current depth k. If $n = k$, the variable refers to the binder currently being opened and is replaced by the free name $\mathrm{free}(x)$. Otherwise, the index refers to a different binder and is left unchanged.

Crucially, when the opening operation encounters an existing free variable $\mathrm{free}(y)$, it simply ignores it. This guarantees that opening is entirely capture-avoiding by construction. In the operational semantics, when a single binder like an input prefix $z(\bullet).P$ receives a channel x, the continuation P is opened as $\{0 \mapsto x\}P$. For a restriction $\boldsymbol{\nu}(\bullet,\bullet)\,P$, the dual endpoints x and y are opened sequentially as $\{0 \mapsto x\}(\{1 \mapsto y\}P)$.

Closing (Abstraction). Closing is the exact inverse of opening. It is the operation of taking an open term and abstracting a specific free name x into a bound index at depth k, denoted $\{k \leftarrow\!\!\shortmid x\}P$. While recursing, bound indices are ignored as they cannot contain free names. When the recursion encounters a free term like $\mathrm{free}(y)$, it checked if $y = x$. If they match, the free name is replaced by $\mathrm{bound}(k)$.

This operation is primarily used when constructing new binders, such as moving a free channel into a restricted scope. To close multiple free names simultaneously, such as abstracting x and y into a restriction, the operations are applied sequentially, by first closing y at depth $k+1$, followed by closing x at depth k.

Locally Closed Terms. The separation of names and indices introduces a new syntactic anomaly, namely that a term can contain a *dangling* index that does not point to any enclosing binder. For example, the process term $\mathrm{bound}(0)[\,].\mathbf{0}$ is syntactically valid, matching the syntax for an empty send, but mathematically it is meaningless because the index 0 has no corresponding binder.

To filter out these meaningless terms, the locally nameless architecture introduces the predicate of being *locally closed*, denoted $lc(P)$. A term is locally closed if and only if all of its De Bruijn indices are properly bound by an enclosing constructor.

The locally closed predicate excludes terms containing dangling indices and therefore characterises the well-formed syntax of the calculus. Establishing this predicate is not only a static validation step, but rather it forms a key invariant throughout the mechanisation, which guarantees the soundness of the calculus. Generally, any subsequent definition, such as variable substitution, structural congruence, or operational semantics, must be mathematically proven to *preserve* the locally closed property. In practice, this is achieved through the *regularity of typing*, which requires proving that all valid typing derivations inherently produce locally closed terms. Consequently, operations proven to preserve typing will naturally carry this invariant as a structural guarantee.

3.4 Co-finite Quantification

With the syntax partitioned into names and indices, and the mechanics of opening and closing established, the final theoretical hurdle is defining how inductive relations, such as typing judgements and labelled transitions, operate across binders.

In paper-based proofs, when a rule requires opening a binder, Barendregt's variable convention allows the author to simply state: *pick a fresh name x*. In a proof assistant, this informal generation of a fresh name must be encoded into the constructors of the inductive relations. Two naive approaches to bind this fresh name, both of which introduce severe mechanical flaws [Aydemir et al. 2008; Charguéraud 2012]:

(1) **Existential Quantification ($\exists x \notin \mathrm{f}(P)$):** The rule requires the property to hold for at least one fresh name. While this makes proof construction straightforward, it yields weak induction hypotheses, since induction only provides the property for one particular fresh name chosen during the original derivation.
(2) **Universal Quantification ($\forall x$):** The rule requires the property to hold for every name. This provides a strong induction hypothesis, but is too restrictive for proof construction,

since the property cannot generally be shown for names that already occur free in the surrounding context.

To solve this dilemma, the locally nameless architecture employs **co-finite quantification** [Aydemir et al. 2008; Charguéraud 2012]. Rather than quantifying over one name or all names, the inductive rules quantifies universally over all names *except* those not in an arbitrary, finite set L. Formally, a rule requiring a fresh name uses the premise: $\forall x \notin L$.

By deferring the exact choice of the forbidden set L to the user at the time of proof, co-finite quantification achieves the best of both worlds. When constructing a proof tree, the user can instantiate L as the set of all free variables currently in the environment, process, and transition labels ($\mathrm{f}(P) \cup \mathrm{f}(\Gamma) \cup \dots$). This ensures the premise is trivially satisfiable. Conversely, when performing structural induction, the co-finite universal quantifier guarantees that the induction hypothesis applies to *any* fresh name the user could possibly need, provided they pick one outside the finite set L.

Application to πLL Types. Although type variables are represented using the locally nameless separation of free and bound constructors, their metatheory is handled using depth-indexed reasoning rather than cofinite quantification. Because type variables are static and do not dynamically communicate across parallel scopes like channels do, managing freshness sets L for them introduces unnecessary proof overhead.

Instead, the system evaluates polymorphic typing rules using a purely depth-indexed approach, using a De Bruijn shifting operator. We define the shifting of a type A by a distance d at a cutoff depth c, denoted mathematically as $\uparrow_d^c A$. Using this infrastructure, consider the typing rule for the universal type input ($\forall.B$):

$$\frac{n + 1 \vdash P :: \uparrow_0^1 \Gamma, x : B}{n \vdash x(\diamond).P :: \Gamma, x : \forall.B}\ \forall_{\mathrm{DB}}$$

Rather than opening the binder with a fresh explicit name, the typing judgment simply increments a depth counter $n + 1$, legally bringing index 0 into scope. To maintain correct index distances, the typing environment Γ is explicitly shifted by a distance of 1 at cutoff 0 ($\uparrow_0^1 \Gamma$), incrementing all free type indices within the context to prevent accidental capture.

Application to πLL Channels. This co-finite quantification strategy fundamentally reshapes the inductive definitions for process channels. For example, consider the typing rule that introduces the prefix $x(\bullet).P$. Under a paper-based presentation, the premise implicitly requires a fresh session name y. Under our mechanised architecture, the rule uses co-finite quantification and the opening operation:

$$\frac{\forall y \notin L, \quad n \vdash \{0 \mapsto y\}P :: \Gamma, x : B, y : A}{n \vdash x(\bullet).P :: \Gamma, x : A \parr B}\ \parr_{\mathrm{LN}}$$

Here, the rule states that the process is well-typed if, for any fresh channel variable y outside some finite set of forbidden variables L, the opened process P is well-typed under the environment extended with the opened payload. By utilizing this mixed evaluation strategy, co-finite quantification for dynamic channels, and depth-indexed evaluation for static types, the architecture minimizes bureaucratic renaming overhead while maintaining a unified underlying syntax.

With this mathematical foundation established, the conceptual theory of πLL is fully prepared for implementation. The following chapter details how this locally nameless architecture is concretely mechanised in the Lean 4 proof assistant.

4 Formalisation

This section describes the mechanisation of πLL in Lean 4, bridging the theoretical foundations developed in Sections 2 and 3 and their concrete implementation. The formalisation is organised into three top-level namespaces: `PiLL.Model`, which defines the static structure of the calculus, `PiLL.Semantics`, which defines its operational behaviour, and `PiLL.SessionFidelity`, which establishes the metatheoretic results connecting the two. Each component is presented in dependency order, highlighting design decisions and divergences from Montesi and Peressotti [2021].

4.1 Model

The `Model` namespace formalizes the static structure of the full πLL calculus and is organized around the layered architecture introduced in Section 2. The mechanisation proceeds bottom-up: session types, defined in `STypes/`, provide the binding structure used by processes in `Processes/`, processes are then typed relative to environments and hyperenvironments from `Environments/` and `HyperEnvironment/`, finally, typing derivations and their metatheoretic infrastructure are collected in `Judgement/`. This includes inversion lemmas, linearity properties, substitution principles, and the co-finite freshness machinery described in Section 3.4. Shared syntactic infrastructure used across these layers is provided by `Base.lean`.

4.1.1 Session Types. Session types are defined in `STypes/Basic.lean` as the inductive datatype `Types`. Following the locally nameless syntax established in Section 3.2, type variables are isolated into a separate inductive type, `TVar`, which explicitly distinguishes between free and bound occurrences:

```
inductive TVar : Type
  | free  (x : FTVar)
  | bound (x : BTVar)
```

Here, `FTVar` represents a free type variable identified by a natural number, while `BTVar` represents a bound type variable encoded as a De Bruijn index.

The remaining constructors directly mirror the theoretical grammar of the calculus. In addition, the formalisation introduces uninterpreted atomic types (`atom` and `atomDual`), representing protocol leaves not further decomposed by the session type grammar and providing the base cases for the inductive structure of the syntax. Table 3 illustrates a representative correspondence between the paper syntax, the locally nameless representation, and the Lean implementation.

Table 3. Comparison of standard, locally nameless, and Lean representations for πLL types.

Standard	Locally Nameless	Lean Constructor	Binds
$A \otimes B$	$A \otimes B$	`| tensor (A B : Types)`	None
1	1	`| one`	None
$\forall X.A$	$\forall.A$	`| forall_ (A : Types)`	1 type
$\exists X.A$	$\exists.A$	`| exists_ (A : Types)`	1 type
X	X	`| var (v : TVar)`	None

Notably, the polymorphic binders $\forall.A$ and $\exists.A$ reflect the depth-indexed approach detailed in Section 3.4. The constructors keep the bound De Bruijn index implicit, taking only the continuation

body A as an argument (an underscore "_" has been appended to the constructor names to prevent namespace collisions with Lean's built-in `forall` and `exists` keywords).

Duality. `Types.dual` implements duality as a structurally recursive function, mapping each constructor to its dual. For example:

```
def Types.dual : Types -> Types
  | .tensor A B  => .parr    (dual A) (dual B)
  | .forall_ A   => .exists_ (dual A)
  | .one         => .bot
  | ...
```

The fundamental property that duality is an involution, $(A^\perp)^\perp = A$, is proved in `STypes/Lemmas.lean` by structural induction.

The formalisation additionally establishes that duality preserves local closure and commutes with both shifting and substitution:

$$A.\mathtt{lc} \iff A^\perp.\mathtt{lc} \qquad \uparrow_d^c (A^\perp) = (\uparrow_d^c A)^\perp \qquad B^\perp\{A/k\} = (B\{A/k\})^\perp$$

Local closure. `STypes/Properties.lean` defines local closure for types as a depth-indexed predicate `Types.lc (k : Nat) : Types → Prop`, where the depth parameter k tracks how many binders the recursion has traversed.

At the leaves of the syntax tree, local closure is evaluated by the helper predicate `TVar.lc`. Because free variables are explicit they do not contain any indices, as such they are inherently locally closed at any depth. By contrast, a bound variable is only locally closed if its De Bruijn index i is strictly less than the current depth k, ensuring it points to an enclosing scope:

```
def TVar.lc : Nat -> TVar -> Prop
  | _, .free _   => True
  | k, .bound i  => i < k
```

The primary `Types.lc` predicate is then defined by structural recursion. For non-binding constructors, the depth parameter distributes across the sub-terms. When a variable is encountered, the predicate delegates the check to `TVar.lc`. When descending into the body of a polymorphic binder, the depth counter increments by one, bringing the newly introduced index safely into scope:

```
def Types.lc : Nat -> Types -> Prop
  | _, .one          => True
  | k, .var v        => TVar.lc k v
  | k, .tensor A B   => A.lc k ∧ B.lc k
  | k, .forall_ A    => A.lc (k + 1)
  | k, .exists_ A    => A.lc (k + 1)
  | ...
```

Finally, a type is defined as *closed* if it contains no dangling indices at the top level, meaning it is locally closed at depth 0. This is registered as the abbreviation `Types.lc_0`.

Shifting. `STypes/Shift.lean` defines shifting following the standard De Bruijn shifting operation introduced in Section 3.4. The operation traverses the syntax tree structurally using a cutoff depth d and a shift amount c.

For standard connectives, shifting distributes recursively across subterms, while base constructors such as one and bot remain unchanged. When traversing a polymorphic binder like `forall_`, the cutoff depth increments by one to account for the newly entered scope:

```
def Types.shift (d c : Nat) : Types -> Types
  | .var v       => .var (v.shift d c)
  | .tensor A B  => .tensor (A.shift d c) (B.shift d c)
  | .forall_ A   => .forall_ (A.shift (d + 1) c)
  | t            => t
```

At the variable level, free variables remain unchanged. For bound indices, the function compares the De Bruijn index i against the cutoff depth d. Indices satisfying $i < d$ are locally bound within the traversed scope and therefore remain unchanged. Otherwise, the index is incremented by c to preserve its reference to the surrounding environment:

```
def TVar.shift (d c : Nat) : TVar -> TVar
  | .bound i => if i < d then .bound i else .bound (i + c)
  | .free n  => .free n
```

Substitution. `STypes/Subst.lean` defines substitution as the replacement of a target De Bruijn index k in a host type B by a payload type A, denoted $B\{A//k\}$.At the variable level, the function compares a bound De Bruijn index i with the target depth k. If $i = k$, the target index has been reached and the variable is replaced by A. If $i < k$, the variable is bound by an inner scope and is left unchanged. Finally, if $i > k$, the index refers to an outer binder beyond the substituted variable. Since substitution removes the binder corresponding to k, such indices are decremented by one to preserve their relative distance to the surrounding environment.

```
def Types.subst (B A : Types) (k : Nat) : Types :=
  match B with
  | .var (.bound i) =>
      if i == k then A
      else if i > k then .var (.bound (i - 1))
      else .var (.bound i)
```

The operation is lifted to the full `Types` syntax by structural recursion. For ordinary connectives, substitution distributes across subterms. When descending into a polymorphic binder such as `forall_`, the target depth is incremented to $k + 1$, and the payload type A is shifted by one. This prevents the free indices in A from being captured by the newly entered binder:

```
  | .forall_ B => .forall_ (B.subst (shift 0 1 A) (k + 1))
```

Notation. `STypes/Notation.lean` introduces notation mirroring the mathematical presentation from Section 2.3. The only divergence is that exponentials are written $!!A$ and $??A$ to avoid clashes with Lean syntax.

The full inductive definition of `Types` and all lemmas from `STypes/Lemmas.lean` are listed in **??**.

4.1.2 Processes. Processes are defined in `Processes/Basic.lean` as the inductive type `Proc`. Following the established locally nameless architecture, channel references are partitioned into a dedicated `Channel` inductive. Explicit free channels are instantiated as concrete natural numbers (`FPName`). While the raw process syntax permits arbitrary reuse of these numbers, using `Nat` allows the formalisation to systematically compute strictly fresh names by incrementing the supremum of any finite set of existing names (mechanised in `Processes/Fresh.lean`). Conversely, bound channels are represented using De Bruijn indices, encoding their relative distance to the corresponding binder.

```
inductive Channel : Type where
  | free  (x : FPName)
  | bound (x : BPName)
```

As discussed in Section 3.2, binding constructs in the process syntax no longer carry explicit payload variables. Instead, bound occurrences are captured purely by their relative position. Comparing the standard paper grammar with the intermediate locally nameless notation and the final constructors of the `inductive Proc` type illustrates this abstraction:

Standard	Locally Nameless	Lean Constructor	Binds
$x[y].P$	$x[\bullet].P$	`| tensor (x : Channel) (P : Proc)`	1 channel
$x(y).P$	$x(\bullet).P$	`| parr (x : Channel) (P : Proc)`	1 channel
$\nu xy\,P$	$\nu(\bullet,\bullet)\,P$	`| cut (P : Proc)`	2 channels
$x(X).P$	$x(\diamond).P$	`| input (x : Channel) (P : Proc)`	1 type
$x[\text{DUP}](x').P$	$x[\text{DUP}](\bullet).P$	`| duplicate (x : Channel) (P : Proc)`	1 channel

Notice that `cut` takes only a continuation P with no explicit channel arguments, as both dual endpoints are simultaneously bound as indices 0 and 1 within the scope of P. Similarly, for constructs like `tensor`, `parr`, and `input`, the subject channel x over which the interaction occurs is kept explicit, while the transmitted payload is anonymized into the continuation.

Opening and closing. `Processes/OpenClose.lean` defines opening and closing operations following the locally nameless principles established in Section 3.3. In the theoretical presentation, opening a process at depth k with x is denoted as $\{k \mapsto x\}P$, and conversely, abstracting a specific name back into a bound index is denoted as $\{k \leftarrow\!\!\shortmid x\}P$.

To mechanise this in Lean, we denote explicit free channels as $\#x$ (mapping to `Channel.free x`) and introduce custom notation for these substitution operations. Expressions such as $P((k|\#x))$ and $P((k|\#x,\#y))$ denote the opening of a process by instantiating the k-th (and $k+1$-th) bound indices with free channels. Conversely, $P\langle\langle k|\#x\rangle\rangle$ and $P\langle\langle k|\#x,\#y\rangle\rangle$ denote the closing of a process by abstracting x and y back into De Bruijn indices. For convenience, the depth parameter can be omitted when operating at the innermost binder ($k=0$), simplifying the notation to $P((\#x,\#y))$ and $P\langle\langle\#x,\#y\rangle\rangle$.

Because binders are represented positionally, the depth parameter k increments according to the number of channels bound by each constructor. For standard communication prefixes like `tensor`, `parr`, and the server `duplicate` operation, exactly one channel is bound, incrementing k by one. The restriction constructor (`cut`) simultaneously binds two dual endpoints, requiring k to increment by two. Notably, the `input` constructor binds a type variable rather than a channel. Consequently, it does not affect the channel depth parameter k.

```
def Proc.open (P : Proc) (u : Channel) (k : Nat) : Proc :=
  match P with
  | .tensor x P     => .tensor (x.open u k) (P.open u (k + 1))
  | .parr   x P     => .parr   (x.open u k) (P.open u (k + 1))
  | .cut      P     => .cut                 (P.open u (k + 2))
  | .input  x P     => .input  (x.open u k) (P.open u k)
  | ...
```

At the leaves of the syntax tree, this recursive traversal delegates to the base case defined by `Channel.open`. For bound channels, the function evaluates the De Bruijn index i against the target

depth k, replacing it with the instantiated channel v upon an exact match. Explicit free channels remain entirely unaffected:

```
def Channel.open (u v : Channel) (k : Nat) : Channel :=
  match u with
  | bound i => if i == k then v else bound i
  | c => c
```

For restriction the HasOpenTwo typeclass from `Base.lean` is instantiated by opening sequentially: first replacing index $k+1$ with the second endpoint v, then replacing index k with the first endpoint u:

```
instance : HasOpenTwo Proc Channel Channel Nat where
  open_ P u v k := (Proc.open (Proc.open P v (k + 1)) u k)
```

To complete the bidirectional infrastructure, `Proc.close` and `Channel.close` define the corresponding closing operation. Used primarily when abstracting explicit free names into new scopes, `Proc.close` traverses the process tree using a structural recursion identical to `Proc.open`, applying the exact same depth-incrementing rules for single and double binders. When the recursion reaches a leaf node, it delegates to `Channel.close`. If the leaf is a free channel (`FPName`), the function compares it against the target name x, upon a match, the explicit name is *abstracted* into a bound De Bruijn index at the accumulated depth k:

```
def Channel.close (u : Channel) (x : FPName) (k : Nat) : Channel :=
  match u with
  | .free name => if x == name then .bound k else .free name
  | .bound i   => .bound i
```

Local closure. `Processes/LocalClosure.lean` defines local closure for processes as a two-parameter predicate `Proc.lc (k n : Nat) : Proc → Prop`. Because the calculus features a two-sorted syntax, the predicate must maintain independent depth counters: k tracks the depth of channel binders, while n tracks the depth of type binders. The type depth is needed when a process embeds a type, for instance as the payload of an output prefix, since the embedded type must be checked relative to the current type-binder depth.

The structural recursion evaluates each constructor by incrementing the appropriate depth counter according to the number and sort of binders introduced. Channel binders increase the channel depth k, while type binders increase the type depth n. For example, `tensor` binds a single continuation channel, incrementing the depth to $k+1$, whereas `cut` binds two dual endpoints simultaneously, incrementing the depth to $k+2$. Conversely, `input` binds a type variable, leaving k unchanged while incrementing n to $n+1$. Constructors introducing no binders, leave both counters unchanged:

```
def Proc.lc : Nat -> Nat -> Proc -> Prop
  | _, _, .nil          => True
  | k, n, .one x P      => x.lc k ∧ P.lc k n
  | k, n, .tensor x P   => x.lc k ∧ P.lc (k + 1) n
  | k, n, .cut P        => P.lc (k + 2) n
  | k, n, .input x P    => x.lc k ∧ P.lc k (n + 1)
  | ...
```

A process is considered globally *closed* when it contains no dangling indices of either sort, meaning `Proc.lc 0 0` holds. This is registered as the abbreviation `Proc.lc_0`.

Free names and freshness. `Processes/Names.lean` defines `Proc.f : Proc → Finset FPName` to compute the free channel names of a process. The function traverses the syntax tree structurally, collecting all explicit free channel identifiers into a finite set. Since locally nameless binders are represented using anonymous indices, bound channels contribute nothing to this set.

For constructors carrying explicit subject channels, such as `tensor` and `link`, the free names of those channels are combined with the recursively computed free names of their continuations. Constructors without explicit channel identifiers of their own, such as `cut`, contribute only the free names recursively obtained from their continuations:

```
def Proc.f : Proc -> Finset FPName
  | .cut P      => P.f
  | .par P Q    => P.f ∪ Q.f
  | .link x y   => x.f ∪ y.f
  | .tensor x P => x.f ∪ P.f
  | .one x P    => x.f ∪ P.f
  | .nil        => {}
  | ...
```

The resulting finite set is used by `Processes/Fresh.lean` to generate names outside a forbidden context. Given a finite set L of forbidden names, a fresh name is computed as $\max(L) + 1$. The lemmas `exists_one_fresh` and `exists_two_fresh` guarantee one or two distinct names outside L, respectively. These lemmas provide witnesses for the co-finite quantification premises used throughout the typing and transition rules, in particular for restriction, where two fresh and distinct endpoints must be introduced simultaneously.

Substitution. The process syntax supports two substitution operations, reflecting its two-sorted binding structure. Name substitution is defined in `Processes/SubstNames.lean` as the function `Proc.substNames (R T : FPName) : Proc → Proc`, replacing all explicit free occurrences of the target name T with the replacement name R. Since bound channels are represented by De Bruijn indices, name substitution only acts on explicit free channel leaves and leaves binders untouched. Consequently, the operation is capture-avoiding by construction and does not require α-renaming or additional freshness side conditions.

Type substitution is defined in `Processes/SubstTypes.lean` as `Proc.substTypes (A : Types) (k : Nat) : Proc → Proc`, substituting A for the De Bruijn type index k throughout a process. Whenever the traversal encounters an embedded type, such as the payload of an output prefix, it delegates to the type-level substitution operation `Types.subst`. When crossing an `input` prefix, which binds a type variable, the target depth is incremented and the substituted type is shifted, mirroring the binder case of `Types.subst` from Section 4.1.1:

```
def Proc.substTypes (A : Types) (k : Nat) : Proc -> Proc
  | .input x P => .input x (P.substTypes (A.shift 0 1) (k + 1))
```

A consequence of this representation is that no analogous channel shifting operation, such as `shiftNames`, is required. In a pure De Bruijn encoding, variables are represented as relative indices and must be shifted when crossing binders. Free channels, however, are represented as global identifiers rather than relative positions, so their representation is unaffected by binder depth. This avoids structural adjustment of channel names during substitution and simplifies the process-level infrastructure.

Structural Properties. `Processes/Lemmas.lean` establishes the fundamental structural properties required by the later typing and operational metatheory. This includes commutation lemmas for

interacting operations (e.g., `open_substNames_comm`, showing that opening commutes with name substitution), preservation properties for local closure (e.g., `lc_of_open`), and lemmas relating binders to free-name analysis (e.g., `f_close`, proving that closing removes a free name from the resulting term).

Of particular importance are the commutation lemmas between opening and substitution: `open_substNames_comm` establishes that opening and name substitution commute, and the corresponding lemma for type substitution ensures that substituting under a binder produces the same result regardless of order. These properties are essential in the typability proof, where the co-finite witnesses produced by the typing rules must be reconciled with the concrete names introduced by the process transitions.

These results provide the core infrastructure required for the transition semantics developed in subsequent chapters.

For completeness, the full inductive definition of `Proc` and all lemmas from `Processes/Lemmas.lean` are listed in **??**.

4.1.3 Environments. Environments are defined in `Environment/Basic.lean` as `Env`, an abbreviation for a list of channel-type pairs:

```
abbrev Elem := (FPName × Types)
abbrev Env  := List Elem
```

The choice of `List` as the underlying representation deserves explanation. Theoretical environments are unordered and duplicate-free, suggesting `Finset` as a natural implementation. However, since `Finset` is implemented as a quotient over lists, it complicates structural induction, pattern matching, and inversion over environments. This proved inconvenient in a development where typing derivations are frequently inverted and typing environments must be rearranged explicitly. A second attempt using association lists (`AList`) encountered similar proof-engineering difficulties. The final design therefore uses plain lists and recovers the unordered structure through an explicit permutation relation, trading quotient reasoning for structural induction and simpler interaction with Lean's tactic machinery.

Recovering the PCM structure. Merging two environments is implemented as list concatenation, making it a simple structural, but total, function:

```
abbrev Env.merge (Γ Δ : Env) : Env := Γ ++ Δ
infixl:69 "," => Env.merge
```

Because lists do not naturally form a Partial Commutative Monoid (PCM) but are associative by definition, we only need to formally recover commutativity.

Since list concatenation is associative definitionally but not commutative, the commutative structure of environments is recovered by reasoning up to permutation. This is achieved using Lean's built-in `List.Perm` relation, instantiated through a custom `HasPerm` typeclass and denoted by the infix relation $\Gamma \sim \Delta$.

The essential monoidal laws, including commutativity, associativity, and left and right identity for $\varnothing$,[1] are established as:

```
lemma Env.merge_comm (Γ Δ : Env) : Γ,Δ ~ Δ,Γ
lemma Env.merge_assoc (Γ Δ Ξ : Env) : Γ,Δ,Ξ = Γ,(Δ,Ξ)
```

[1]Using $\varnothing$ rather than the literal `[]` requires explicit rewrite lemmas for terms such as $\varnothing, \Gamma$. If `[]` were used directly, $[\,], \Gamma$ would reduce by definitional equality, since `Env.merge` abbreviates `List.append`.

The partiality of environment merging is enforced by explicit predicates rather than by making `Env.merge` a partial function. `Env.names` extracts the domain as a `Finset`, enabling Lean's built-in `Disjoint` typeclass. Internal linearity (no channel typed twice) is enforced by `Env.Nodup`:

```
def Env.names (Γ : Env) : Finset FPName := (Γ.map Prod.fst).toFinset
def Env.disjoint (Γ Δ : Env) : Prop := Disjoint Γ.names Δ.names
def Env.Nodup (Γ : Env) : Prop := (Γ.map Prod.fst).Nodup
```

The relationship between linearity, disjointness, and merging of typing environments is captured by `Env.Nodup_merge_iff`, which states that a merged environment is linear if and only if both component environments are individually linear and mutually disjoint.

Server usability. The typing rule for server replication requires all dependency channels to carry exponential request types. This side condition is captured by `Env.serverUsable`, which asserts that every type in an environment satisfies `Types.isServerUsable`, i.e. has the form $?B$.

```
def Env.serverUsable (Γ : Env) : Prop :=
  ∀p, p ∈ Γ -> p.snd.isServerUsable
prefix:max "?ₑ" => Env.serverUsable
```

Lifted operations. Since environments contain types, the locally nameless operations from Section 4.1.1 are lifted pointwise over the type component of each `Elem`. Local closure, type shifting, name substitution, and type substitution are all defined by mapping over the list. The file `Environment/Lemmas.lean` proves that these operations preserve the names, disjointness, nodup, and permutation properties of environments.

The full definitions and lemmas are listed in **??**.

4.1.4 Hyperenvironments. Hyperenvironments are defined in `HyperEnvironment/Basic.lean` as `HyperEnv`, an abbreviation for a list of environments:

```
abbrev HyperEnv := List Env
```

As for environments, merging is implemented as list concatenation, using $\varnothing$ as the unit and $\mathcal{G}|_h\mathcal{H}$ as notation for composition. Associativity and unit laws therefore follow from the underlying list structure. However, recovering the commutative structure requires more than the ordinary list permutation relation, since hyperenvironments are lists whose elements are themselves environments considered modulo permutation.

Deep permutation. For environments, Lean's `List.Perm` relation is sufficient, since it allows arbitrary reordering of channel-type pairs. For hyperenvironments, however, permutation must operate at two levels: it must allow reordering of component environments, and it must also respect permutation equivalence inside each component environment.

For example, consider the hyperenvironment containing a single component environment $[x : A, y : B]$. Ordinary `List.Perm` treats this component environment as an opaque element, so it does not identify it with $[y : B, x : A]$. The custom relation `HyperEnv.Perm` instead permits component environments to be replaced by permutation-equivalent environments.

```
inductive HyperEnv.Perm : HyperEnv -> HyperEnv -> Prop where
  | nil  : Perm [] []
  | cons : {Γ Δ : Env} {𝒢 ℋ : HyperEnv} : Γ ~ Δ -> Perm 𝒢 ℋ -> Perm (Γ :: 𝒢) (Δ :: ℋ)
  | swap : {Γ Δ : Env} {𝒢 : HyperEnv} : Perm (Γ :: Δ :: 𝒢) (Δ :: Γ :: 𝒢)
  | trans {𝒢 ℋ 𝒥 : HyperEnv} : Perm 𝒢 ℋ -> Perm ℋ 𝒥 -> Perm 𝒢 𝒥
```

The key constructor is cons. Unlike ordinary list permutation, it does not require the head environments to be syntactically identical, it only requires them to be permutation-equivalent. Thus HyperEnv.Perm is a deep permutation relation, propagating equivalence both across the ordering of component environments and within the environments themselves.

Well-formedness. Hyperenvironment well-formedness is decomposed into two predicates. The predicate HyperEnv.Nodup requires each component environment to be internally linear, while HyperEnv.PairwiseDisjoint requires distinct component environments to have disjoint domains:

```
def HyperEnv.Nodup (𝒢 : HyperEnv) : Prop :=
  ∀ Γ, Γ ∈ 𝒢 -> Env.Nodup Γ

def HyperEnv.PairwiseDisjoint (𝒢 : HyperEnv) : Prop :=
  List.Pairwise Env.disjoint 𝒢
```

The distinction is important because the two predicates rule out different violations of linearity. For example, assuming x, y, and z are distinct names:

- $[x : A] \mid_h [y : B, z : C]$ satisfies both predicates.
- $[x : A, x : B] \mid_h [y : C]$ violates Nodup, since x appears twice within one component environment.
- $[x : A] \mid_h [x : B, y : C]$ violates PairwiseDisjoint, since x appears in two distinct components.

Together, these predicates enforce global linearity: every channel appears at most once across the whole hyperenvironment.

Lifted operations. As with environments, type shifting, name substitution, and type substitution are lifted pointwise over the list of component environments. The file HyperEnvironment/Lemmas/Basic.lean proves that these operations preserve names, disjointness, pairwise disjointness, and the deep permutation relation. For example, preservation of permutation under shifting is proved by induction on HyperEnv.Perm, using the corresponding environment-level permutation lemma in the cons case.

The full definitions and lemmas are listed in **??**.

4.1.5 Typing Judgements. The typing relation is defined in Judgement/Basic.lean as the inductive predicate Typing:

```
inductive Typing : Nat -> Proc -> HyperEnv -> Prop
```

We write $n \vdash P :: \mathcal{G}$ for a derivation of this predicate. The index n tracks the current type-binder depth, while the remaining indices record the process being typed and its associated typing environment. This intrinsic indexing is later used by the erasure maps in the semantics and metatheory.

Type-depth indexing. The type-depth index is relevant whenever the derivation crosses a polymorphic binder. For example, the forall_ constructor increments the depth from n to $n + 1$ and shifts the surrounding environment to keep embedded type indices in scope:

```
| forall_ :
    Typing (n + 1) P [x:B :: Γ⁺] ->
    Typing n (#x((◇)).P) [x:∀.B :: Γ]
```

The shift Γ^+ abbreviates shifting the type indices in Γ by one, mirroring the binder-sensitive shifting used by type substitution in Section 4.1.1.

Exchange and explicit side conditions. Because environments and hyperenvironments are represented as lists, the unordered and partial structure of the theoretical hyperenvironment is recovered explicitly. The typing relation includes exchange rules for reasoning up to permutation equivalence:

```
| exchange_env   : Typing n P (Γ :: 𝒢) -> Γ ~ Δ -> Typing n P (Δ :: 𝒢)
| exchange_hyper : Typing n P 𝒢 -> 𝒢 ~ ℋ -> Typing n P ℋ
```

The exchange rules recover the unordered character of the paper presentation without quotienting environments or hyperenvironments. The complementary issue is partiality: although merging is implemented as total list concatenation, typing rules carry explicit disjointness and freshness premises to ensure that only valid linear compositions are formed. For example, `mix` requires the two hyperenvironments to be disjoint, while prefix rules such as `tensor` and `parr` require freshness premises ensuring that subject channels do not already appear in the continuation environment.

The same design principle reappears later in the operational semantics, where transitions must also respect permutation equivalence of environments and hyperenvironments.

Co-finite quantification. Binding rules use co-finite quantification as described in Section 3.4. For example, the `parr` constructor types an input prefix by requiring the opened continuation to be well typed for every fresh payload name outside a finite set L:

```
| parr (hF : x ∉ Γ.names) (L : Finset FPName) :
    (∀y ∉ L, Typing n (P((#y))) [y:A :: x:B :: Γ]) ->
    Typing n (x((•)).P) [x:A ⅋ B :: Γ]
```

Here, the process carries no explicit payload name; the bound occurrence in the continuation is instantiated by opening P with the fresh name y. The premise `hF` enforces that the subject channel x is not already present in the continuation environment. Rules binding multiple names, such as `cut`, follow the same pattern but quantify over two distinct fresh names.

Structural invariants. `Judgement/Names.lean` establishes the central name invariant for the typing relation, named `Typing_f_eq_names`. It states that, for any well-typed process, the free names of the process coincide exactly with the domain of its hyperenvironment.

$$n \vdash P :: \mathcal{G} \implies P.fn = \mathcal{G}.names$$

This ensures exact correspondence between process free names and typed environment entries: no free process channel is left untyped, and no environment entry is unused.

Building on this, `Judgement/Linearity.lean` proves `Typing_preserves_disjointness`: any well-typed hyperenvironment is pairwise disjoint. Together these results guarantee that typing derivations can only be constructed for well-formed, linear contexts.

The full inductive definition of `Typing` is listed in **??**.

4.2 Semantics

The dynamic behaviour of the πMLL fragment is formalised in the `Semantics/` namespace. Following the erasure structure described in Section 2.5, the development contains three related transition systems: a typed transition system over typing derivations, and two projected systems describing the induced process and environment behaviour.

- **`TypingStep`$_{\mathtt{m}}$**: the primary, resource-aware LTS over typing derivations (`JudgementStep/Basic.lean`);
- **`ProcStep`$_{\mathtt{m}}$**: the process-level LTS obtained by forgetting typing resources (`ProcStep/Basic.lean`);

- **EnvStep$_m$**: the resource-level LTS describing the corresponding evolution of hyperenvironments (`EnvStep/Basic.lean`).

Labels. Transition labels are formalised in `Labels.lean`. Although the operational semantics mechanised here target πMLL, the underlying action vocabulary is defined for the full πLL calculus to support future extensions.

The core action type, `Act`, captures observable communication events, including empty communication, channel transmission, type instantiation, and server interaction. Selection and server actions are parameterised by the auxiliary datatype `Mu`. The `Lbl` inductive then extends observable actions with the silent transition τ, explicit session-link labels, and a parallel constructor combining two observable actions directly, reflecting the synchronized action labels used by the parallel rules.

```
inductive Act : Type
  | one       (x : FPName)
  | tensor    (x y : FPName)
  | output    (x : FPName) (A : Types)
  | muBrack   (x : FPName) (μ : Mu)
  | ...

inductive Lbl : Type
  | tau
  | link  (x y : FPName)
  | act   (p : Act)
  | par   (l l' : Act)
```

Each label defines computable sets of free names (`Lbl.f`) and introduced names (`Lbl.i`), which are used in the side conditions for the rules PAR$_1$, SYN and PAR$_2$. To preserve alignment with the theoretical presentation, we use the `HasBracket` and `HasParen` typeclasses for `Act` and `Lbl` to allow the notation for process prefixes to be reused uniformly for labels as well. For instance, $x[[y]]$ denotes both the output process prefix and the corresponding output label.

4.2.1 The Typed Transition System. The primary LTS, `TypingStep`$_m$, is a relation between typing derivations. Its constructors mirror the operational rules shown in Figure 2, but make the type-theoretic content of each step explicit: every transition carries a typing derivation as its source and produces a typing derivation as its target. The system is therefore intrinsically typed.

```
inductive TypingStepₘ :
  {n : Nat} → {𝒢 : HyperEnv} → {P : Proc} → Typing n P 𝒢 →
  Lbl → {n' : Nat} → {𝒢' : HyperEnv} → {P' : Proc} → Typing n' P' 𝒢' → Prop
```

As established previously, both environment and hyperenvironment composition are formalised using total list appends. Consequently, the explicit disjointness proofs required by the static `Typing` relation to safely mix environments must propagate directly into the dynamic `TypingStep`$_m$ relation. For instance, the `par`$_1$ constructor requires explicit hypotheses (`hD1` and `hD2`) to guarantee that the hyperenvironments remain strictly disjoint both before and after the transition:

```
| par₁
    (hD1 : 𝒢.disjoint ℋ) (hD2 : 𝒢'.disjoint ℋ)
    (h : TypingStepₘ 𝒟 l 𝒟')
    (disj : l.i ∩ Q.f = ∅) :
    TypingStepₘ
      (Typing.mix hD1 𝒟 ℰ)
      l
```

```
    (Typing.mix hD2 𝒟' ℰ)
```

The post-state disjointness proof `hD2` is included explicitly as a convenience: while it is derivable from `hD1`, the freshness condition $i(l) \cap Q.f = \emptyset$, and the structural property that transitions only introduce names recorded by their label, keeping it as a premise avoids reconstructing this fact repeatedly in later proofs.

Remark 4.1. Strictly stepping typing derivations can expose structural artifacts that are usually hidden in the paper presentation. For example, in Figure 3, the intermediate derivation is typed by $\varnothing \mid y : 1$ rather than simply $y : 1$, because the MIX rule preserves the empty component contributed by the terminated left process. In Lean, this corresponds to a `Typing.mix` constructor containing a `Typing.mix_∅` sub-derivation. The result is valid, but reducing it to the canonical form requires applying the monoidal identity laws of the hyperenvironment explicitly, rather than by definitional equality. The final $\equiv$ step in the figure, which additionally uses process congruence to identify $\mathbf{0} \mid \mathbf{0}$ with $\mathbf{0}$, is not currently automated because structural congruence has not been mechanised. This limitation is discussed further in Section 6.

4.2.2 Projected Transition Systems. The process and environment transition systems are defined to capture the two erasure views of a typed transition. $\texttt{ProcStep}_{\mathtt{m}}$ records the process behaviour obtained by forgetting typing resources, while $\texttt{EnvStep}_{\mathtt{m}}$ records the corresponding resource evolution. These relations are later connected to $\texttt{TypingStep}_{\mathtt{m}}$ by the simulation theorems in Section 5.1.

Process steps. $\texttt{ProcStep}_{\mathtt{m}}$ models the syntactic evolution of processes independently of typing resources. Unlike the typing relation, it is not indexed by a typing derivation and therefore does not use co-finite premises. Binder-opening rules instead choose concrete fresh names directly and record the required freshness as explicit side conditions. For example, the `tensor` rule opens the continuation with a concrete name y, requiring y to be fresh for both the subject channel and the continuation:

```
| tensor {P : Proc} {x y : FPName} (hF : y ∉ {x} ∪ P.f) :
    ProcStep_m (#x[[•]].P) (x[[y]]) P(#y)
```

This creates a mismatch with the co-finite formulation of the typing rules. Thus $\texttt{ProcStep}_{\mathtt{m}}$ remains close to a standard concrete transition relation, while the additional freshness reconciliation is handled in the typability proof. A co-finite formulation of $\texttt{ProcStep}_{\mathtt{m}}$ would align more directly with the typing rules, but the present formulation keeps the erased LTS closer to a standard concrete transition relation.

Environment steps. The structural evolution of hyperenvironments, written $\mathcal{G} \xrightarrow{l} \mathcal{H}$, is formalised by $\texttt{EnvStep}_{\mathtt{m}}$. Unlike the typed transition system, $\texttt{EnvStep}_{\mathtt{m}}$ deliberately contains only the resource transformation described by the label. It does not independently enforce all disjointness and well-formedness conditions. Those properties are supplied by the typing derivation in $\texttt{TypingStep}_{\mathtt{m}}$.

This separation keeps the projected resource semantics lightweight. The environment relation describes how resources change, while the typed semantics proves that these changes occur from well-formed linear contexts.

The following metatheory shows that these transition systems agree: typed transitions project to process and environment transitions, and process transitions from well-typed states can be lifted back to typed transitions.

5 Metatheory

This section establishes the metatheoretic results connecting the typed, process, and environment transition systems defined in Section 4.2. The goal is to verify that the formalisation is sound: well-typed processes evolve according to their session types, and their associated linear resources evolve consistently. The results are contained in the `SessionFidelity/` namespace and cover the multiplicative fragment πMLL.

The paper additionally develops behavioural theory including bisimilarity, the diamond property, serialisation, non-interference, and readiness [Montesi and Peressotti 2021]. These results are not mechanised in the current formalisation and are discussed as future work in Section 7.

5.1 Session Fidelity

Having defined the typed, process, and environment transition systems, we now establish the metatheoretic results connecting them. The goal is to verify Session Fidelity for the mechanised πMLL fragment: well-typed processes evolve according to their session types, and their associated linear resources evolve consistently with the process transition.

The formalisation, contained in the `SessionFidelity/` namespace, proceeds in two directions. First, erasure simulation shows that typed derivation steps project to valid process and environment transitions. Second, typability and subject reduction show that process transitions from well-typed processes can be lifted back to typed derivation steps and therefore preserve typing.

Erasure simulations. The paper establishes erasure as a strong bisimulation between $lts_d$and lts_e (Theorem 4.7 in Montesi and Peressotti [2021]), meaning the relation $\{(proc(\mathcal{D}), \mathcal{D}) \mid \mathcal{D} \in Der\}$ is closed in both directions. The mechanisation currently establishes only the forward direction via `session_fidelity_proc`$_m$. The backward direction, which would require lifting any process step to a typed step, is subsumed by `typability_subject_reduction`$_m$ but has not been composed into an explicit bisimulation statement.

The files `ProcStep.lean` and `EnvStep.lean` establish that typed derivation steps simulate correctly under erasure. Given a typed transition `TypingStep`$_m$ $\mathcal{D}$ l $\mathcal{D}'$, the process extracted from the source derivation transitions to the process extracted from the target derivation, and similarly for the associated environment.

```
theorem session_fidelity_proc_m
  {𝒟 : Typing n P 𝒢} {𝒟' : Typing n' P' 𝒢'}
  (hStep : TypingStep_m 𝒟 l 𝒟') :
  ProcStep_m (proc 𝒟) l (proc 𝒟')
```

```
theorem session_fidelity_env_m
  {𝒟 : Typing n P 𝒢} {𝒟' : Typing n' P' 𝒢'}
  (hStep : TypingStep_m 𝒟 l 𝒟') :
  EnvStep_m (env 𝒟) l (env 𝒟')
```

These results establish that the typed transition system refines both projected semantics: every typed step has a corresponding process step and a corresponding environment step under erasure. The proofs proceed by induction on the typed transition derivation.

Typability. The main lifting theorem is mechanised in `Typability.lean`. It states that an untyped process transition from a well-typed process can be lifted to a transition between typing derivations:

```
theorem typability_subject_reduction_m
  (𝒟 : Typing n P 𝒢) (hPS : ProcStep_m P l P') :
  ∃ (𝒢' : HyperEnv) (𝒟' : Typing n P' 𝒢'),
```

```
    TypingStepₘ 𝒟 l 𝒟'
```

This theorem is the technical core of the session fidelity proof. It is proved by induction on the process transition derivation. Each case inverts the source typing derivation to recover the typing rule compatible with the observed process shape, then constructs a target hyperenvironment and typing derivation for the resulting process.

The main mechanisation burden comes from reconciling the concrete freshness witnesses used by $\texttt{ProcStep}_\texttt{m}$ with the co-finite premises of the typing rules, and from rearranging hyperenvironments in the communication and restriction cases. In particular, synchronization under restriction requires isolating the environments containing the communicating endpoints, applying the transition under opened names, and reconstructing the target hyperenvironment up to permutation.

Subject reduction. Subject reduction is stated in `SubjectReduction.lean`. It is the standard type-preservation theorem for the process semantics: if a well-typed process takes a process transition, then there exists a corresponding evolved hyperenvironment under which the target process is well typed.

```
theorem subject_reductionₘ
  {n : Nat} {𝒢 : HyperEnv} {P P' : Proc} {l : Lbl}
  (𝒟 : Typing n P 𝒢) (hPS : ProcStepₘ P l P') :
  ∃ 𝒢', EnvStepₘ 𝒢 l 𝒢' ∧ Typing n P' 𝒢'
```

Mechanically, this theorem is obtained by composing typability with environment erasure. First, $\texttt{typability\_subject\_reduction}_\texttt{m}$ lifts the process step to a typed derivation step, producing a target typing derivation. Then $\texttt{session\_fidelity\_env}_\texttt{m}$ projects this typed step to an environment transition. Thus the process and its typing resources evolve coherently under the same label.

Implementation considerations. Compared with the paper presentation, the formalisation makes several implicit structural obligations explicit. Since environment and hyperenvironment composition are implemented by total list concatenation, disjointness and freshness conditions appear as explicit premises in the typing and transition rules. Moreover, the process LTS uses concrete freshness witnesses, while the typing rules use co-finite quantification. This mismatch does not change the mathematical content of the semantics, but it introduces additional proof obligations in the typability theorem.

Interpretation. Together, these results establish the intended session fidelity property for the mechanised fragment. The process semantics cannot move independently of the typing environment: every process step from a well-typed source has a corresponding resource transition, and the resulting process remains well typed under the evolved hyperenvironment. Conversely, typed derivation steps project faithfully to both process-level and environment-level behaviour.

The paper additionally establishes behavioural properties including bisimilarity, the diamond property, serialisation, non-interference, and readiness for πMLL. These results are not mechanised in the current formalisation; extending the mechanisation to cover them is discussed in Section 7.

6 Discussion

This section reflects on the design choices made throughout the formalisation, discusses alternatives that were explored, and identifies limitations of the current mechanisation.

Design evolution of session fidelity. The proof of session fidelity underwent significant revision before reaching its current form. An early iteration of the formalisation attempted to prove typability by induction on the typing derivation rather than on the process transition. This approach stalled

because inverting a typing derivation for a parallel process exposes two independent sub-derivations, but the process transition only steps one side, leaving the proof with no straightforward way to reassemble the target derivation. The final proof instead inducts on $\texttt{ProcStep}_{\mathtt{m}}$ and inverts the typing derivation in each case, which gives direct access to the specific typing rule that produced the observed process shape.

An earlier version of the session fidelity development also involved explicit infrastructure for closing processes and associated commutation lemmas, as well as auxiliary lemmas for the duplication and disposal constructors. As the formalisation matured and the scope was restricted to πMLL, much of this infrastructure was removed or simplified. The current development is accordingly leaner, though extending to the full πLL calculus would require revisiting some of this earlier work.

An earlier approach to the typability proof attempted to invert `EnvStep` directly, requiring a substantial set of inversion lemmas (`EnvStep_inv_bot`, `EnvStep_inv_one`, `EnvStep_inv_one_bot`, `EnvStep_inv_link`) together with auxiliary `HyperEnv.Perm.extract_*` lemmas for each communication pattern under restriction. This infrastructure was necessary because the smart `EnvStep` carried enough information to reconstruct the hyperenvironment structure from a transition alone. The current dumb design eliminates this entire layer, since `EnvStep` is a simple projection of `TypingStep`, inversion is handled through the typing derivation instead, significantly reducing the proof burden.

Redundant premises and proof convenience. Several premises in the typing and transition relations are included for proof convenience rather than logical necessity. The most prominent example is the post-state disjointness proof `hD2` in the `par` constructors of $\texttt{TypingStep}_{\mathtt{m}}$, discussed in Section 4.2.1. This premise is derivable from the pre-state disjointness proof `hD1`, the freshness condition $i(l) \cap Q.f = \emptyset$, and the structural property that transitions only introduce names recorded by their labels. Keeping `hD2` explicit avoids reconstructing this derived fact in every inductive case, at the cost of slightly cluttering the inductive definition. Identifying and removing such redundant premises would produce cleaner inductive definitions and is a natural cleanup task alongside the co-finite refactoring.

Co-finite quantification in ProcStep. The current $\texttt{ProcStep}_{\mathtt{m}}$ relation uses concrete freshness witnesses rather than co-finite quantification for the binding constructors `tensor`, `one_bot`, and `res`. This creates a mismatch with the co-finite premises of the typing rules, which introduces additional proof obligations in the typability theorem , specifically, the proof must reconcile the concrete names chosen by $\texttt{ProcStep}_{\mathtt{m}}$ with the universally quantified premises of the typing rules. Reformulating $\texttt{ProcStep}_{\mathtt{m}}$ to use co-finite quantification for these constructors would make both relations more uniform and could simplify the typability proof. Whether this simplification is achievable without complicating the erasure simulation proofs remains to be verified.

Structural congruence. The formalisation does not include a structural congruence relation for processes. In the paper presentation of πLL, structural congruence identifies processes that differ only by the commutative, associative, and unit laws of parallel composition, as well as scope extrusion for restriction. The monoidal laws for hyperenvironments established in Section 4.1.4 capture the typing-level counterpart of these laws, and the exchange rules in the typing relation recover commutativity at the derivation level. However, a full process-level congruence relation, and the associated proof that well-typed processes are closed under it, has not been mechanised. This would require showing, for example, that $P \mid \mathbf{0}$ and P are interchangeable in any typing derivation, which currently requires manual application of the monoidal identity lemmas at each step.

An alternative approach suggested by the structure of the formalisation would be to handle associativity via a label-swapping rule in the transition system rather than in structural congruence:

$$\frac{\nu xy\, P \xrightarrow{\mu} P'}{\nu yx\, P \xrightarrow{\mu} P'}$$

This would avoid duplicating all typing rules and could handle scope symmetry without a separate congruence relation. Whether this approach is compatible with the current induction structure of the session fidelity proofs remains to be investigated.

Limitations of the current mechanisation. The formalisation covers the full static structure of πLL but restricts the operational semantics and metatheory to πMLL. Several results from the paper metatheory remain unmechanised, including the behavioural theory (bisimilarity and congruence), the diamond property, serialisation, non-interference, and readiness. These are discussed as directions for future work in the following section.

Proof automation. A recurring cost throughout the formalisation is the manual rearrangement of hyperenvironments using permutation and disjointness lemmas. Many proof steps reduce to showing that two hyperenvironments are equivalent up to permutation and satisfy disjointness, which currently requires explicit sequences of rewrite steps. Building a dedicated `aesop` rule set equipped with the hyperenvironment permutation and disjointness lemmas could automate much of this framing work and significantly reduce proof size.

Incomplete cases in typability. The `typability_subject_reduction`$_{\mathtt{m}}$ proof is complete for the structural cases but admits several cases involving binder-opening via `sorry`. The affected cases are those requiring simultaneous freshness witnesses for `res` and `tensor_parr`, where the existential freshness conditions of `ProcStep`$_{\mathtt{m}}$ must be reconciled with the co-finite premises of the corresponding typing rules. The proof obligation is understood and structurally clear; the remaining work is the explicit construction of the required permutation and renaming lemmas. These cases are discussed further in Section 7.

7 Future Work

The most immediate directions for future work concern completing the operational semantics and extending the formalisation to cover the full πLL calculus.

Completing the multiplicative fragment. The current session fidelity proofs cover πMLL with the exception of the restriction rule. Completing this case would require additional reasoning about how restriction interacts with itself under synchronisation, and how the associated hyperenvironment components are preserved and rearranged across steps. The hyperenvironment permutation infrastructure already developed for the other cases provides the necessary foundation; the remaining work is primarily in constructing the specific extraction lemmas for the restriction-under-restriction scenario.

Co-finite quantification in ProcStep. The current `ProcStep`$_{\mathtt{m}}$ relation uses existential freshness conditions for binding constructors, while the typing rules use co-finite quantification. This asymmetry introduces proof overhead in the typability theorem, where the specific concrete names chosen by `ProcStep`$_{\mathtt{m}}$ must be reconciled with the universally quantified premises of the corresponding typing rules. Refactoring `ProcStep`$_{\mathtt{m}}$ to use co-finite quantification for its binding constructors, specifically `tensor`, `one_bot`, and `tensor_parr`, would eliminate this mismatch, remove the `all_fresh` helper infrastructure, and likely allow the post-state disjointness premise `hD2` to be dropped entirely. Whether this refactoring also resolves the remaining `sorry` cases

in `typability_subject_reduction`$_{\mathtt{m}}$ is the primary open question and the main motivation for pursuing this direction.

Extending to full πLL. Once the multiplicative fragment is complete, extending the formalisation to the full πLL calculus would require reinstating the step relation constructors for the additive, exponential, and polymorphic fragments. A prior revision of the formalisation included these constructors together with the auxiliary `buildDup` and `buildDisp` process constructions needed for the duplication and disposal rules. The static model, processes, types, environments, and typing judgements, is already defined for the full calculus, so the remaining work is concentrated in the operational semantics and session fidelity proofs.

Structural congruence. Mechanising structural congruence would address the artificial stuck states that arise from the strict binary tree structure of the process syntax. The most practically significant laws are commutativity and associativity of parallel composition, which are needed for the SYN rule to apply to processes that are not immediate siblings, as discussed in Section 6. Scope extrusion introduces additional complexity in the locally nameless setting, since pulling a process under a restriction binder requires explicit De Bruijn shifting. Rather than adding a full congruence relation to the process syntax, a more tractable approach may be to extend the transition system with explicit symmetric structural rules, such as a `syn-comm` constructor, which would allow transitions modulo the relevant congruences while preserving the syntactic stability needed for proof search in Lean.

Behavioural metatheory. The paper establishes a range of further results that are not mechanised in the current formalisation, including bisimilarity and congruence (Theorems 4.5, 4.9, 4.10 in Montesi and Peressotti [2021]), the diamond property (Theorem 4.15), serialisation (Fact 4.18), non-interference (Theorem 4.20), and readiness (Theorem 4.21). These results build directly on session fidelity and would become accessible once the session fidelity proofs are complete.

Proof automation. A recurring cost throughout the formalisation is the manual rearrangement of hyperenvironments using permutation and disjointness lemmas. Building a dedicated `aesop` rule set equipped with the hyperenvironment permutation and disjointness lemmas could automate much of this framing work and significantly reduce proof size across the development.

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